



CS17L Software

Requirement Specification

The University of Mindanao

College of Computing Education

Bachelor of Science in Computer Science

by

Joanne Faith N. Cabarde

Robroi Neo M. Dingal

Angelo O. Lozano

School Year: 2025-202

1. INTRODUCTION

1.1 Purpose

The purpose of this Software Requirement Specification (SRS) is to document the comprehensive requirements for the RN Milaran Hardware Management System (RNM-HMS). This document outlines the functional and non-functional requirements, user roles, and system scope to guide the development team and ensure all business operations (sales, inventory, purchasing, and users) are effectively managed.

1.2 Intended Audience

The intended audience for this document includes the project developers, software engineers, testers, and the stakeholders (store owners and administrators of RN Milaran Hardware).

1.3 Intended Use

The system is intended to be used daily by the hardware store's Administrator or Owner. The Administrator will have complete, unrestricted access to the system to oversee all business operations. This includes processing customer transactions through the Point of Sales (POS) module, managing the inventory lifecycle (such as manual stock adjustments and archiving), creating purchase invoices for inbound goods, and maintaining supplier records. Additionally, the Administrator will use the system to manage employee access, monitor system audit logs, and gain overarching business insights from the Dashboard. .

1.4 Product Scope

The RNM-HMS is a centralized web-based application designed to replace manual record-keeping in a hardware retail environment.

Benefits: It eliminates manual errors, speeds up checkout times, provides real-time stock tracking, simplifies supplier purchasing, and secures the business with role-based access and audit trails.

Objectives: To create an integrated system that connects POS checkout directly to inventory deduction, while also linking supplier purchases to inventory additions.

Goals: Improve operational efficiency, prevent stockouts through dashboard alerts, maintain accurate financial records, and streamline the employee workflow.

1.5 Definitions

Point of Sales: The module used for processing customer checkout, modifying item markups, and generating receipts.

Inventory Management System: The module handling stock overview, manual adjustments, and stock movement logs.

Supplier Records System: The module for maintaining supplier contact data and status.

Purchasing System: The module used to generate invoices for restocking goods from suppliers.

User Management System: The module for creating and managing employee accounts and access levels.

Archive: A feature that allows administrators to safely hide products that have exactly zero stock without deleting historical data.

1.6 Overview

The RNM-HMS offers a unified solution for hardware store management. It integrates five core modules which are POS, Inventory, Purchasing, Suppliers, and User Management and supported by a centralized Dashboard and System Audit Logs to ensure total visibility and control over the business.

1. OVERALL DESCRIPTION

2.1 Product Perspective

RN Milaran Hardware Management System (RNM-HMS) is an internal management platform designed to modernize and streamline the operations of RN Milaran Hardware. It integrates core business processes into a centralized system where administrators can manage inventory, process sales through a Point-of-Sales (POS) module, maintain supplier records, handle purchasing for inbound goods, and manage user roles and permissions. By replacing traditional manual pen-and-paper practices, the system aims to improve transaction efficiency, ensure data integrity, and eliminate inefficiencies that can lead to mismanagement.

The system is composed of five primary subsystems: Point of Sales (POS) for real-time transaction handling; Inventory Management for rigorous stock tracking and low-supply notifications; Supplier Records for storing partner information; Purchasing for managing invoices and new product entries; and User & Role Management for controlling system access. Together, these modules provide a professional and well-structured framework that supports accurate financial reporting and operational growth.

The development of the RNM-HMS is planned as a focused project involving key personnel, including Front-End and Back-End developers, a UI/UX Designer, and a Project Manager. The timeline is structured to complete the core functional processes, including full Create, Read, Update, and Delete (CRUD) capabilities across all subsystems, ensuring a fully functional interface for both administrative and operational use.

2.2 User Stories

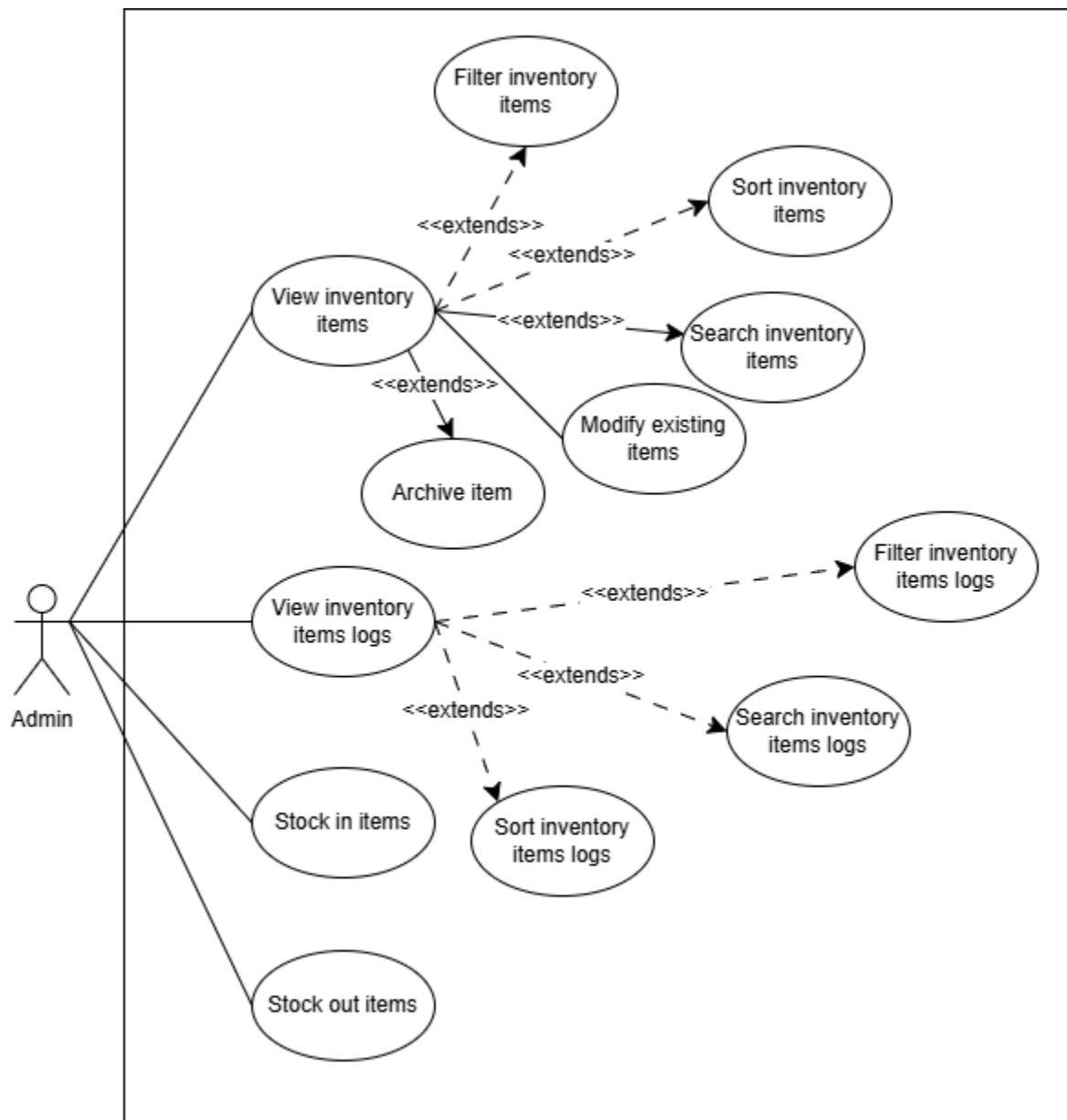
- **As an Admin / Store Owner**, I want to view a dashboard with total sales and low-stock alerts so that I can make quick, informed business decisions.
- **As an Admin**, I want to view system audit logs so that I can track which employee performed specific actions in the system.

2.3 Use Cases

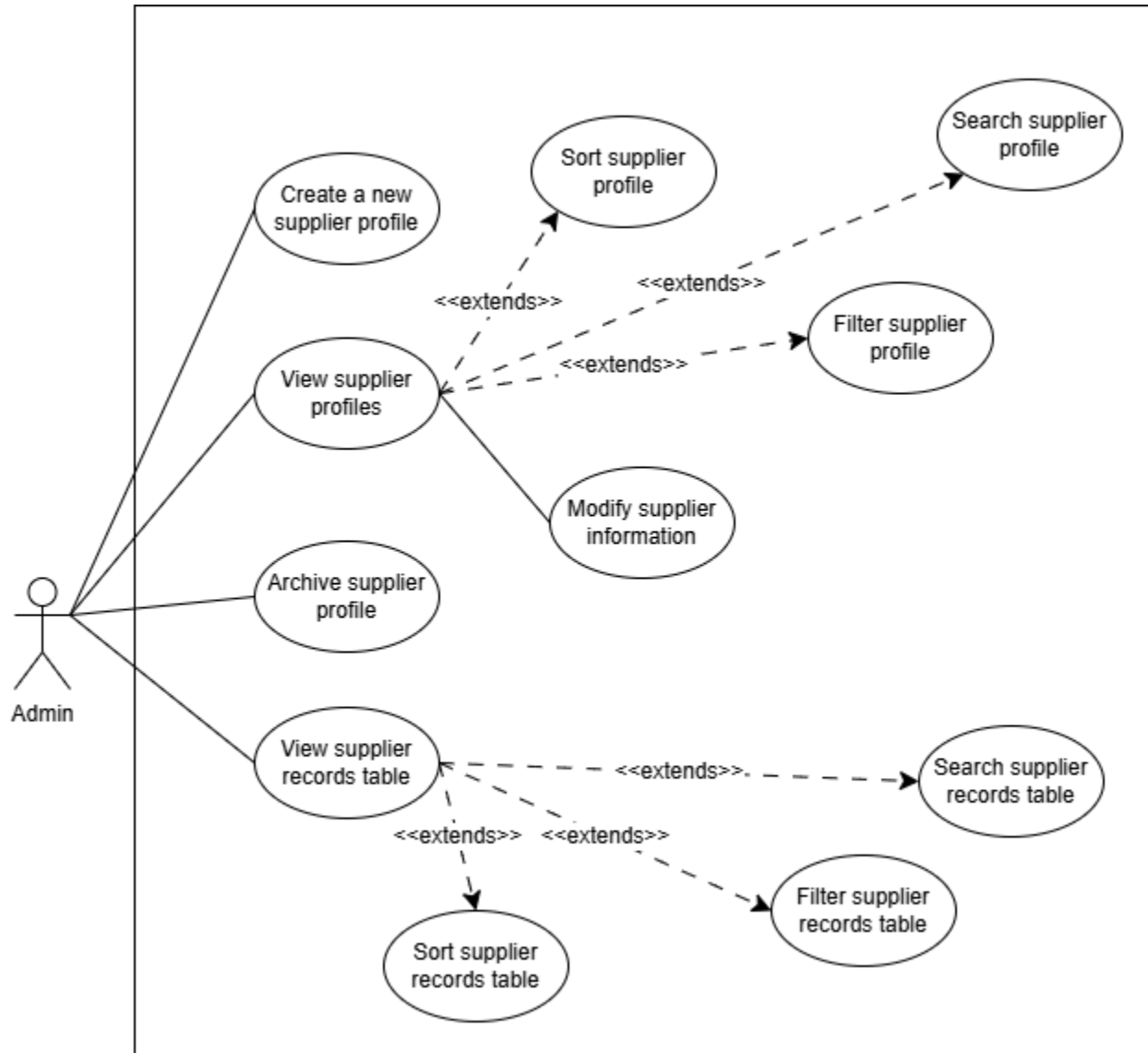
POINT-OF-SALES



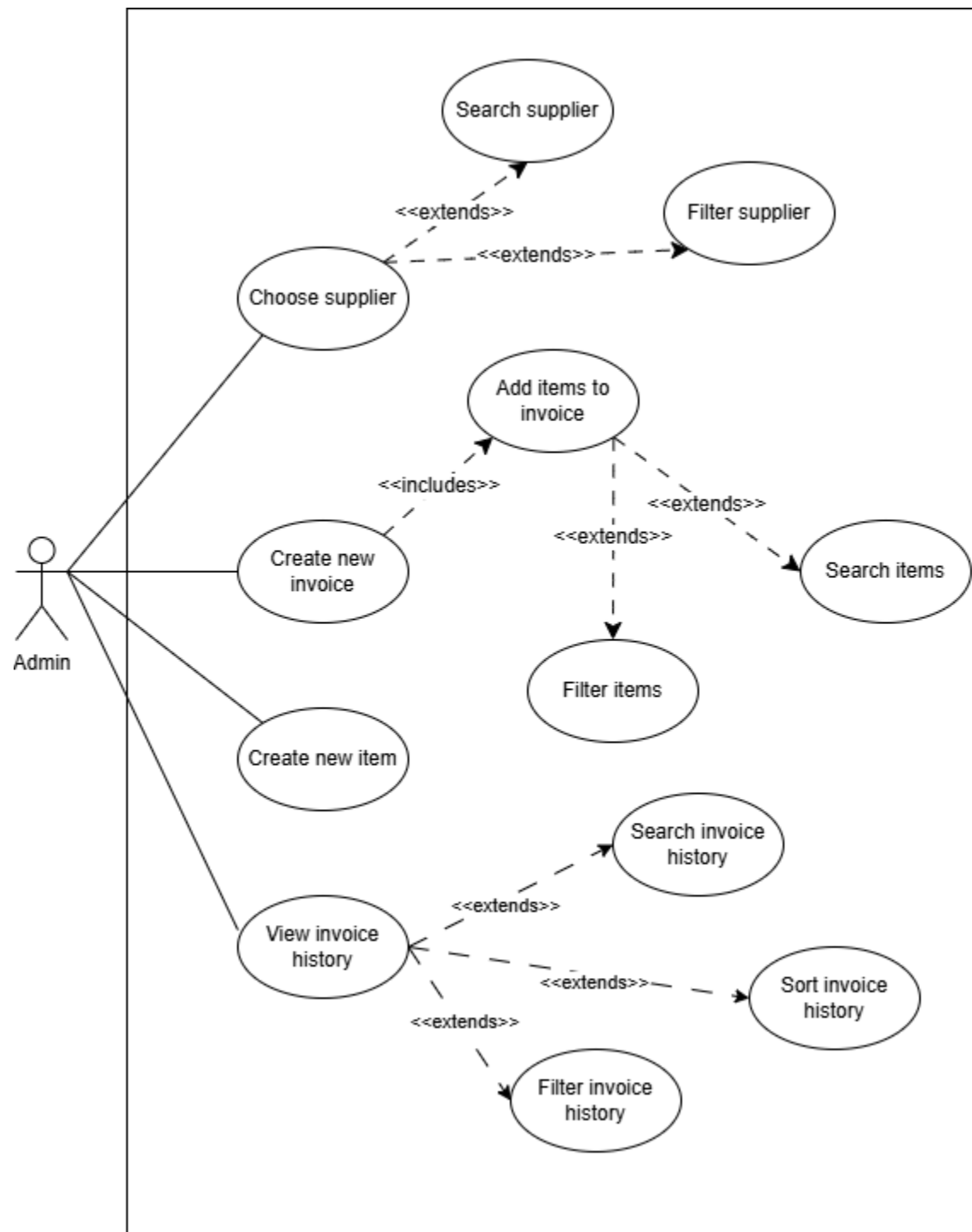
INVENTORY MANAGEMENT SYSTEM



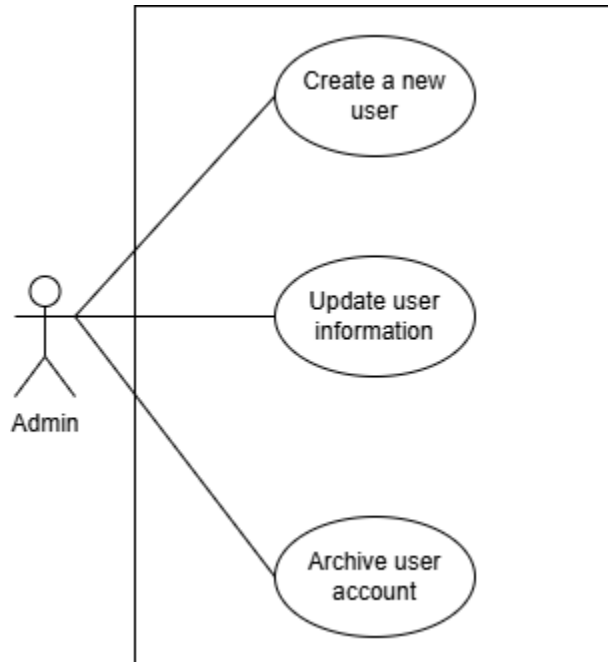
SUPPLIER RECORDS SYSTEM



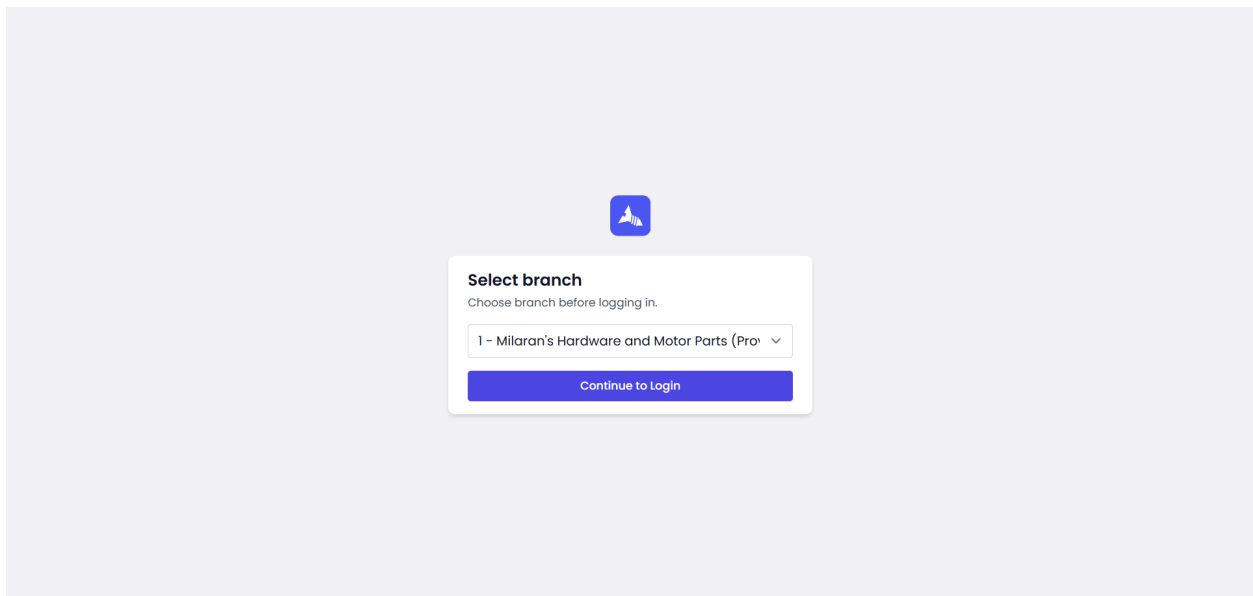
PURCHASING SYSTEM

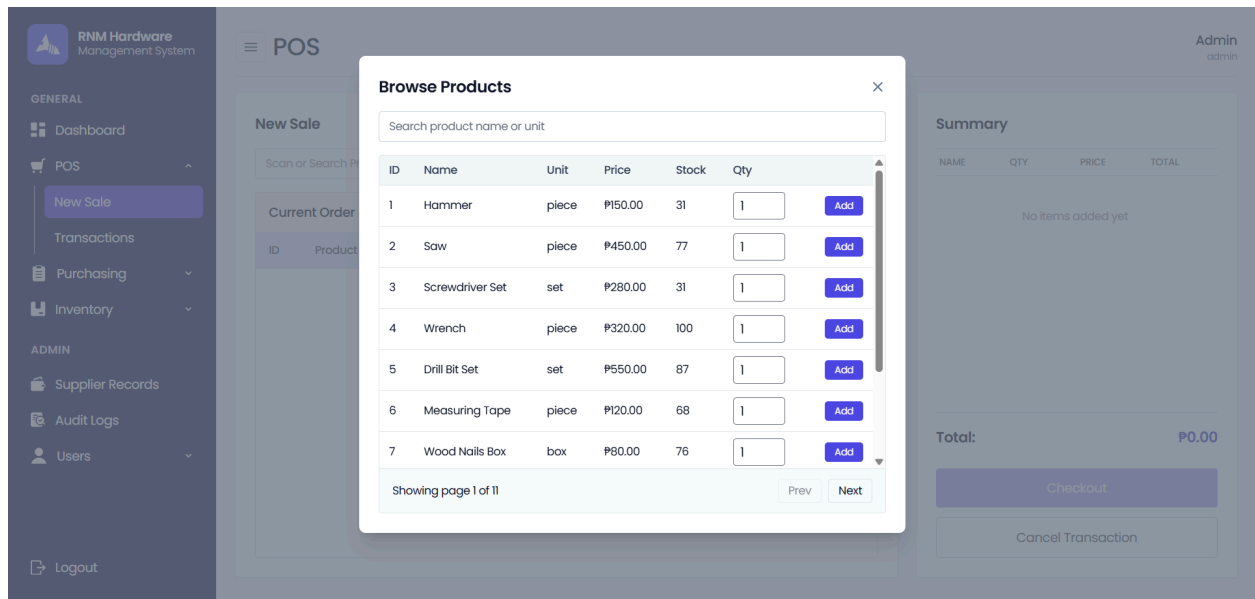
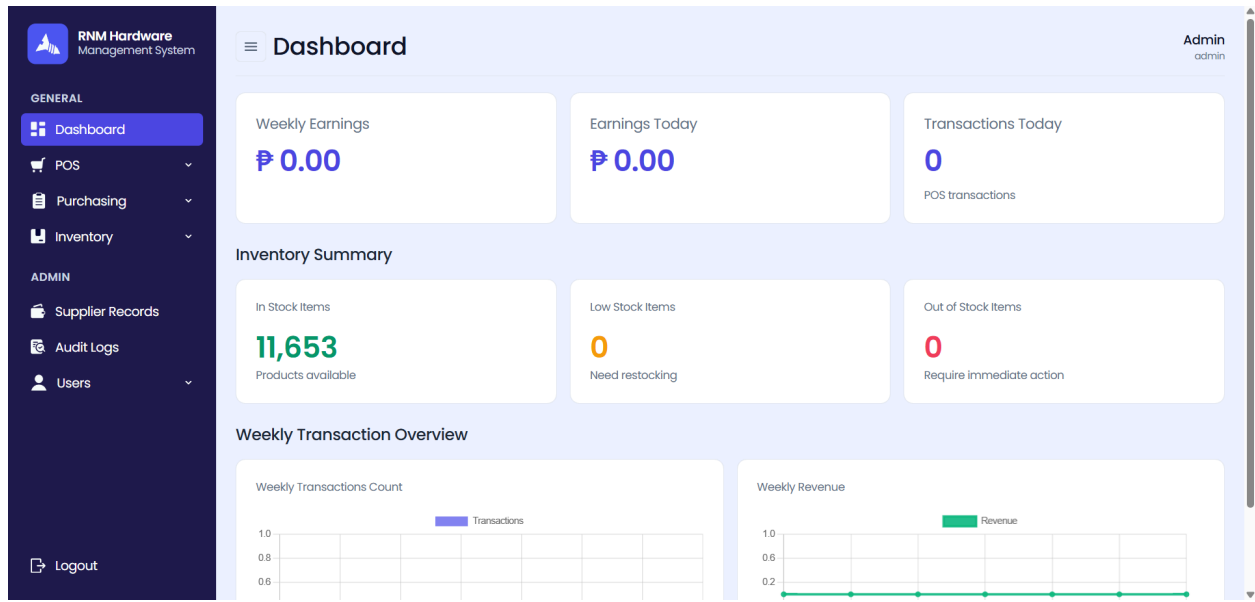


USER AND ROLE MANAGEMENT SYSTEM



2.4 Software Prototypes





POS

Admin
admin

New Sale

Scan or Search Product ID, name, or unit...

Browse Products

Current Order

ID	Product Name	Unit	Capital	Markup	Quantity	Subtotal	
2	Saw	piece	P450.00	P 0	1	P450.00	Remove

Summary

NAME	QTY	PRICE	TOTAL
Saw	1	P450.00	P450.00

Total: P450.00

Checkout

Cancel Transaction

POS

Admin
admin

New Sale

Scan or Search Product ID, name, or unit...

Current Order

ID	Product Name	Unit	Capital
No products in order. Use "Browse Products"			

Receipt

Close

Sale ID: 1

Date: 2026-05-27 21:16:50

Cashier: Admin

Branch: Milaran's Hardware and Motor Parts

Terminal: T -

Payment: CASH

Item	Qty	Price	Subtotal
Saw #2 · piece	1	P450.00	P450.00

Total P450.00

Print Done

Summary

NAME	QTY	PRICE	TOTAL
No items added yet			

Total: P0.00

Checkout

Cancel Transaction

RNM Hardware Management System

GENERAL

Dashboard

POS

Purchasing

New Invoice

Invoice History

Inventory

ADMIN

Supplier Records

Audit Logs

Users

Logout

New Invoice

Adminadmin

Supplier & Branch

Branch

Milaran's Hardware and Motor Parts

Supplier *

Prime Hardware Distributors

Supplier Contact Information

Contact Person

David Chen

Contact Number

09668593142

Email

procurement@primehw.com

Company Address

789 Enterprise Road, Supply City, SC 67890

Invoice Details

Payment Terms

Net 30

Due Date

06/26/2026

Invoice Date

May 27, 2026

Summary

NAME	QTY	PRICE	TOTAL
No items added yet			
Total:			₱0.00
Proceed to Checkout			
Clear Cart			

Add Products

Search or Create Products

Scan or Search Product ID, name, or unit...

+ New Product

Product	Unit	Quantity	Unit Price	Subtotal
No products added. Search above or create a new product to get started.				

RNM Hardware Management System

GENERAL

Dashboard

POS

Purchasing

Inventory

Stock Overview

Manual Stock In

Manual Stock Out

Stock Movements

ADMIN

Supplier Records

Audit Logs

Users

Logout

Stock Overview

Adminadmin

Scan or Search Product ID, name, or unit...

Filter by Branch

All Branches

Filter by Status

All Statuses

Total Products

110

Total Value

₱6,007,195.20

Low Stock Items

0

ID	Product Name	Unit	Quantity	Unit Cost	Total Value	Branch	Status	Action
1	Hammer	piece	31.00	₱150.00	₱4,650.00	Milaran's Hardware and Motor Parts	In Stock	—
2	Saw	piece	76.00	₱450.00	₱34,200.00	Milaran's Hardware and Motor Parts	In Stock	—
3	Screwdriver Set	set	31.00	₱280.00	₱8,680.00	Milaran's Hardware and Motor Parts	In Stock	—
4	Wrench	piece	100.00	₱320.00	₱32,000.00	Milaran's Hardware and Motor Parts	In Stock	—
5	Drill Bit Set	set	87.00	₱550.00	₱47,850.00	Milaran's Hardware and Motor Parts	In Stock	—
6	Measuring Tape	piece	68.00	₱120.00	₱8,160.00	Milaran's Hardware and Motor Parts	In Stock	—
7	Wood Nails Box	box	76.00	₱80.00	₱6,080.00	Milaran's Hardware and Motor Parts	In Stock	—
8	Iron Bolts Pack	pack	49.00	₱200.00	₱9,800.00	Milaran's Hardware and Motor Parts	In Stock	—
9	Pliers	piece	54.00	₱180.00	₱9,720.00	Milaran's Hardware and Motor Parts	In Stock	—

GENERAL

Dashboard

POS

Purchasing

Inventory

ADMIN

Supplier Records

Audit Logs

Users

Logout

Supplier Management

Adminadmin

Suppliers List

+ Add Supplier

Filter by Status

All Statuses

Search suppliers by name or phone...

Search

Clear

Supplier Name	Contact Person	Contact Number	Status	ACTIONS
TechWorks Inc	John Smith	09630147793	Active	
Global Electronics Ltd	Maria Garcia	09801524617	Active	
Prime Hardware Distributors	David Chen	09668593142	Active	
FastShip Components	Sarah Johnson	09649278018	Active	
Reliable Supply Co	Robert Williams	09987894186	Active	
Vintage Tech Supplies	Emily Brown	09530112315	Inactive	

GENERAL

Dashboard

POS

Purchasing

Inventory

ADMIN

Supplier Records

Audit Logs

Users

Logout

New User

Adminadmin

Add New User

BASIC INFO

Full name *

Address *

Contact number *

09123456789

ROLE & ACCESS

Role *

-- Select role --

Branch *

-- Select branch --

PIN

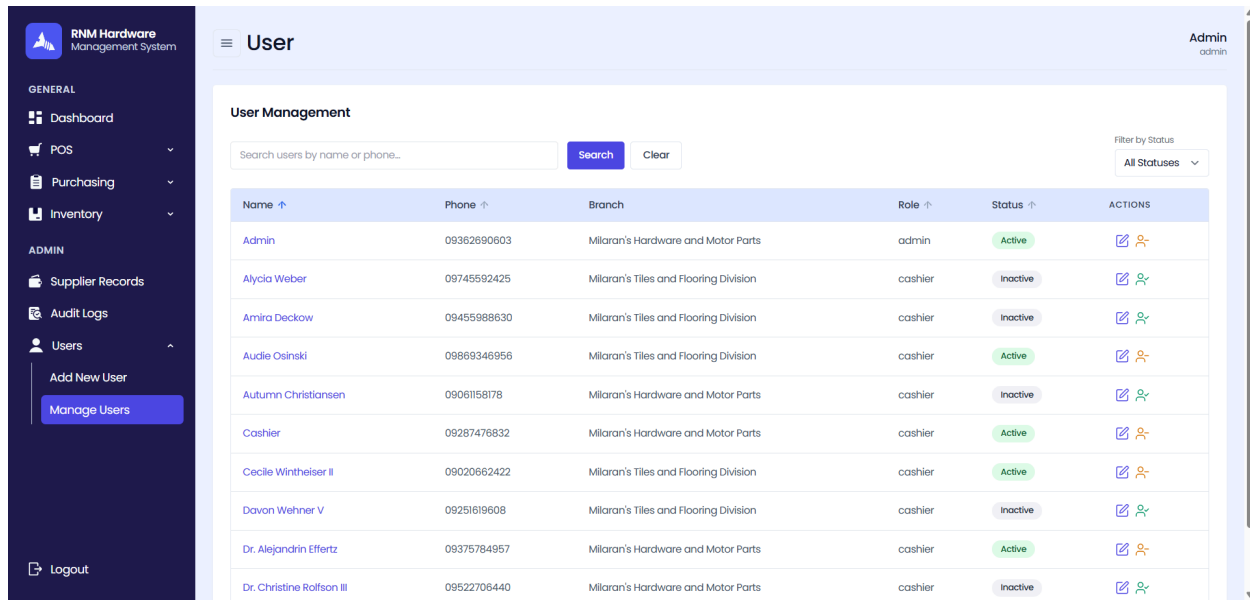
PIN *

Confirm PIN *

The user will use this PIN to log in to the system.

Cancel

Create User



2.5 User Description

Admin:

Role: The Administrator has complete technical and structural responsibility for the platform's maintenance, security, and data integrity.

Authorization: The Admin has unrestricted access to all layers of the system. This includes the User Management Subsystem (creating/updating employee profiles, setting role classifications), viewing the System Audit Logs, and issuing transaction refunds. They hold the highest authorization level, allowing them to archive inventory entries, toggle supplier operational statuses, and modify global software parameters.

2.6 Product Functions

POINT-OF-SALES

Code	F001
Description	Create Sale & Receipt Generation .
Input	Product search query, quantities, modified markups, cash payment, and checkout confirmation.
Processing	Queries database, calculates subtotal with markups, processes payment, updates stock, and writes to sales/sales_items tables.
Output	Saved sales transaction record, updated inventory levels, and payload sent to the thermal printer.
User Interface	POS Checkout Screen, Product Search Component, and Payment

	Settlement Dialog.
Dependencies	Existing Product Details and Checkout Cart

Code	F002
Description	Cart Alteration and Cancellation
Input	"Remove Item" row click or global "Cancel Order" command.
Processing	Deletes targeted item record array from active cache or completely flushes the temporary cart session memory.
Output	Refreshed cart layout showing reduced items or an empty checkout canvas.
User Interface	Action control column in Cart Grid (Trash Icon) and "Cancel Transaction" button.
Dependencies	Active checkout cart

Code	F003
Description	Sales Transaction History
Input	Navigation trigger targeting the Sales History workspace.
Processing	Fetches records from sales table combined with a relational user join to output rows chronologically.
Output	Historical table visualization displaying IDs, timestamps, payment methods, and operator profiles.
User Interface	Sales History Main Registry Panel and Tabular Reporting Grid.
Dependencies	None

Code	F004
Description	The system must allow users to search, sort, and filter the sales history table.
Input	Transaction History Search, Sort, and Filter
Processing	Modifies query statements dynamically using string matches (LIKE),

	date ranges (WHERE), and directional ordering (ORDER BY).
Output	Refined, sorted, or filtered transaction entries displayed inside the reporting grid.
User Interface	Upper Search Strip, Date-pickers, and toggleable column headers.
Dependencies	Existing sales history details

Code	F005
Description	Sales Refund
Input	Target Transaction ID, refund confirmation execution, and authorization prompt.
Processing	Verifies Admin access, updates transaction state to 'Refunded', restores line item values back into inventory stock, and logs action.
Output	Updated invoice status, replenished product quantities, and an appended system audit trace.
User Interface	Transaction Detail Drawer and Refund Confirmation Warning Modal.
Dependencies	Existing sales history details

INVENTORY MANAGEMENT SYSTEM

Code	F001
Description	View and Search Inventory
Input	Alphanumeric keyword (Name/ID), sort parameters, and category filters.
Processing	Pulls active rows from the products table and injects real-time filtering, keyword matching, and pagination filters.
Output	Tabular screen grid populated with matching active stock rows.
User Interface	Inventory Dashboard Grid, Search Bar, and Filter Dropdowns.
Dependencies	Existing Inventory Details

Code	F002
Description	Modify Existing Inventory Item
Input	Target Product ID, revised alphanumeric field inputs, and "Save" click.
Processing	Validates field size rules and executes a database UPDATE statement on the targeted products table row configuration.
Output	Updated database parameter modifications and an on-screen success notification toast.
User Interface	Inventory Grid Row Options and "Edit Product Parameters" Modal Overlay.
Dependencies	Existing Inventory Details

Code	F003
Description	Archive Item
Input	Target Product ID and "Archive" confirmation trigger.
Processing	Verifies QuantityInStock == 0, switches visibility flag to archived, and shifts item snapshot index to the archive_logs partition.
Output	Target item removed from active grid views and saved inside the archived partition ledger.
User Interface	Action control options list and the "Archive Security Validation" Warning Modal.
Dependencies	Targeted product quantity metric must equal exactly 0.

Code	F004
Description	Stock Movement
Input	Automated background hooks containing Product ID, change quantity deltas, transaction type, and user ID.
Processing	Captures runtime mutation event payloads and appends a time-stamped tracking entry line string directly into product_logs.
Output	Secure, unedited ledger trace line added to the master inventory audit history.

User Interface	Stock Movements Ledger Tab Workspace Grid.
Dependencies	None

SUPPLIER RECORDS SYSTEM

Code	F001
Description	Create Supplier
Input	Supplier Name, Contact Representative, Address, and Phone Number.
Processing	Validates input format constraints and executes a database INSERT query into the suppliers table.
Output	New supplier record saved to the database and an on-screen success notification toast.
User Interface	"Add New Supplier Profile" Form Screen / Modal.
Dependencies	Supplier Details

Code	F002
Description	Update Supplier
Input	Target Supplier ID and revised field inputs (Name, Representative, Address, Phone Number).
Processing	Validates structural data edits and executes a database UPDATE statement on the targeted row index.
Output	Revised partner parameters committed to the database and an update success notification.
User Interface	Supplier Details Editing Form Panel or inline modification drawer.
Dependencies	Existing supplier details

Code	F003
Description	Update Supplier Status
Input	Target Supplier ID and Status Toggle interaction click.
Processing	Locates row reference inside suppliers table and flips binary visibility

	flag to hide/show partner across the procurement application.
Output	Immediate operational visibility status state correction.
User Interface	Supplier Registry Management Table showing inline sliding Status Toggle Switches.
Dependencies	Existing supplier details

PURCHASING SYSTEM

Code	F001
Description	Create Invoice
Input	Target Supplier ID, Reference Bill Serial Number, Invoice Date, and purchase item selections.
Processing	Confirms merchant visibility flag is active and opens a transaction session container inside temporary working cache files.
Output	Active inbound purchase worksheet interface mapped to the vendor file.
User Interface	Procurement Workspace Creation Panel, Dropdown Search menus, and Calendar controls.
Dependencies	Existing Supplier and Product details

Code	F002
Description	Create New Item
Input	New Product Title Name, Base Measurement Unit selection, and Initial Capital wholesale cost values.
Processing	Confirms description uniqueness and saves a fresh configuration profile directly to the products table with zero initial base stock.
Output	Active product configuration item row token ready for purchase line attachments.
User Interface	Inline "New Item" Modal Popup window inside Purchasing view.
Dependencies	Product Details

Code	F003
Description	Checkout Invoice
Input	Purchase Line items and checkout trigger execution click.
Processing	Saves totals to invoices table, increments target item metrics inside products table via additions, and dispatches background logs.
Output	Confirmed wholesale invoice layout document and updated item counts across the active inventory module.
User Interface	Procurement Manifest Review Control Drawer and Checkout Submission processing views.
Dependencies	None

USER AND ROLES MANAGEMENT SYSTEM

Code	F001
Description	User Login
Input	Employee Email address string and account password string or keypad PIN code values.
Processing	Hashes input strings, compares them against employers table records, and generates an encrypted role token context baseline.
Output	Workplace navigation access clearance redirection or login credential authentication exception blocks.
User Interface	Unified Application Platform Portal Identity Sign-In Page interface.
Dependencies	Valid user

Code	F002
Description	Manage Users Profile
Input	Legal Name, Security Role selection ('Admin', 'Cashier', 'Manager'), unique corporate email, and new Passphrase string.
Processing	Verifies email uniqueness across the application datastore, cryptographically hashes passwords, and saves records into employers.

Output	Revised user identity and system operational boundary mappings committed to the security tables.
User Interface	Identity Matrix Administration Panel Screen, User Display Tables, and Account Provisioning Forms.
Dependencies	User Details

Code	F003
Description	System Audit Log
Input	Automated data hooks containing execution timestamps, acting user identity tags, target modules, and alphanumeric action details.
Processing	Captures structural platform mutations in real time and updates unalterable activity entries chronologically into the global audit table.
Output	Continuous row records appended to the unyielding master security audit history ledger tracking window.
User Interface	System Operations Master Security Audit Logs Overview Display Panel Grid.
Dependencies	Invoked automatically by any write, update, delete, or status switch executions across all primary modules.

2. FUNCTIONAL REQUIREMENTS

Functional Requirements for Admin Dashboard:

(Users indicated below are: Administrator)

Code	Description	Remarks
FR001	The system must display accurate, real-time dashboard summary metrics (e.g., Total Sales, Total Products, Total Purchases) fetched directly from the database.	None
FR002	The system must display low stock alerts, correctly identifying and listing inventory items that have fallen below the minimum	None

	stock threshold.	
FR003	The system must provide a sidebar navigation menu that successfully redirects users to corresponding modules (POS, Inventory, Supplier Records, Purchasing, Users) without loading errors.	None

Functional Requirements for Point of Sales (POS):

(Users indicated below are: Administrator)

Code	Description	Remarks
FR001	The system must allow users to create a sale by browsing or searching for products, modifying markups, and confirming cash payments to generate a receipt.	None
FR002	The system must allow users to remove selected products from the cart or cancel an ongoing order entirely before checkout.	None
FR003	The system must display a history of previous transactions, including ID, Date, Total Amount, Payment Method, and the processing user.	None
FR004	The system must allow users to search, sort, and filter the sales history table.	None
FR005	The system must allow authorized users to process a refund for an existing transaction and update the transaction logs accordingly.	None

Functional Requirements for Inventory Management System (IMS):

(Users indicated below are: Administrator)

Code	Description	Remarks
FR001	The system must display a stock overview	None

	table detailing product ID, Name, Unit, Quantity, Cost, Total Value, Branch, and Status.	
FR002	The system must allow users to perform manual stock-in and stock-out adjustments and reflect these changes accurately in the system.	None
FR003	The system must maintain a detailed stock movements log displaying Date, Time, Product, Unit, Type of Movement, Quantity Change, Branch, and User.	None
FR004	The system must allow users to search, sort, and filter both the active stock overview and the stock movement logs.	None
FR005	The system must allow users to modify or edit existing product details (e.g., price, unit, name) in the database.	None
FR006	The system must allow authorized users to archive a product only if its current stock quantity is exactly zero.	None
FR007	The system must log archiving actions in the Stock Movements table and allow users to unarchive (restore) items from there.	None

Functional Requirements for Supplier Records System (SRS):

(Users indicated below are: Administrator)

Code	Description	Remarks
FR001	The system must allow users to add new suppliers by inputting relevant contact and business details.	None
FR002	The system must allow users to edit existing supplier details and update the database in real-time.	None
FR003	The system must allow users to	None

	deactivate active suppliers or activate inactive suppliers.	
FR004	The system must allow users to search, sort, and filter the supplier records table by status or keyword.	None
FR005	The system must display a toast notification providing immediate visual feedback upon the success or failure of supplier record actions.	None

Functional Requirements for Purchasing System (PS):

(Users indicated below are: Administrator)

Code	Description	Remarks
FR001	The system must allow users to process a purchase invoice by selecting a supplier, adding products, inputting quantity and price, and checking out.	None
FR002	The system must automatically update the inventory stock levels upon the successful confirmation of a purchase invoice.	None
FR003	The system must allow users to create and register a new product into the system directly from the purchasing module interface.	None
FR004	The system must maintain and display a complete invoice history (ID, Purchase ID, Supplier, Date, Amount, Items).	None
FR005	The system must allow users to search, sort, and filter the purchasing invoice history table.	None

Functional Requirements for User Management System & Auth:

(Users indicated below are: Administrator)

Code	Description	Remarks
FR001	The system must authenticate user credentials (username and PIN/password) via a login interface and deny access for invalid inputs.	None
FR002	The system must enforce role-based access, directing authenticated users to interfaces appropriate for their role (e.g., Admin vs. Cashier).	None
FR003	The system must allow administrators to register new user accounts with specific roles, branches, and contact details.	None
FR004	The system must allow administrators to edit user details, as well as activate or deactivate employee accounts.	None
FR005	The system must allow administrators to search, sort, and filter the manage users table.	None
FR006	The system must maintain and display an Audit Log of all system events, including the date, time, user, action performed, and affected module.	None

3. NON-FUNCTIONAL REQUIREMENTS

4.1. Usability

Intuitive Interfaces: The POS checkout interface and Purchasing invoice flow must feature a clear, logical layout to allow for fast, error-free processing by staff.

Visual Cues: Low stock items within the inventory module must be visually distinct (e.g., color-coded) for quick identification.

Responsive Design: System interfaces, including navigation bars and data tables, must be accessible and scale appropriately across different monitor sizes used in the hardware store.

4.2. Security

Access Control: All dashboard data, audit logs, and inventory archiving features must strictly require administrative privileges. Unauthorized roles must be physically unable to access these modules.

Data Protection: User passwords and PINs must be securely hashed and encrypted in the database, never visible in plain text. Sensitive supplier contact information must also be protected from unauthorized access.

Immutability: Confirmed purchase invoice histories and stock movement logs cannot be altered or tampered with once saved to the database.

4.3. Reliability

Concurrency Handling: The POS and Purchasing systems must accurately handle concurrent transactions and stock adjustments without double-billing, data corruption, or inventory miscalculations.

Accuracy: Calculations for total quantities, markups, costs on invoices, and inventory valuations must be consistently accurate under all conditions.

Safe Deletion: The system must safely retain inactive user and supplier data (soft deletion) without deleting it permanently to preserve historical transaction integrity.

4.4. Performance

Load Times: The dashboard and large data tables (like the complete invoice history or stock overview) must load in less than 2 seconds, even with large datasets.

Search Speed: Searching for a specific product in the POS cart, or filtering a specific supplier from a large database list, should return results instantly (under 1 second).

System Stability: Filtering, sorting, and navigating through extensive audit logs or inventory items must process smoothly without screen freezing or causing the system to time out.

4. EXTERNAL INTERFACE REQUIREMENTS

5.1. User Interfaces

The system is a web-based application designed with a focus on usability, speed, and responsiveness. The frontend is built utilizing Tailwind CSS and Alpine.js to ensure a modern, clean, and interactive user experience.

Navigation: A persistent sidebar navigation menu will be available across all screens for Administrators to easily switch between the Dashboard, POS, Inventory, Suppliers, Purchasing, and Users modules.

POS Interface: The checkout screen will feature a split-screen layout—a product search/browse section on one side and the active cart/invoice on the other—designed for fast-paced data entry by cashiers.

Data Tables: Modules like Inventory, User Management, and Audit Logs will utilize dynamic data tables equipped with built-in search bars, up/down sorting arrows, and filter dropdowns.

System Feedback: The system will utilize "Toaster" pop-up notifications (green for success, red for errors) located at the corner of the screen to provide immediate, non-intrusive feedback for actions like adding suppliers or archiving items.

Modals: Critical or destructive actions, such as archiving an inventory item with 0 stock, will trigger a centered pop-up confirmation modal to prevent accidental clicks.

5.2. Hardware Interfaces

Because the RNM-HMS is a web-based platform, it does not require heavy, proprietary server hardware on-site (if hosted in the cloud), but it does require specific peripheral hardware to fully execute its functional requirements:

Client Computers/Tablets: Standard desktop computers, laptops, or tablets running a modern operating system (Windows, macOS, Linux) with a minimum of 4GB RAM to prevent browser lag during large database queries.

Receipt Printer: The Point of Sales (POS) client computer must be connected to a thermal receipt printer to fulfill the "Print receipt" action upon checkout confirmation.

Cash Drawer: A standard cash register drawer connected via the receipt printer (or USB) to store physical cash payments confirmed during POS checkout.

Barcode Scanner (Optional/Supported): While users can search products manually by typing, the POS and Purchasing modules will support standard plug-and-play USB/Bluetooth barcode scanners functioning as keyboard input devices for faster product lookups.

5.3. Software Interfaces

The application relies on a specific software stack to function securely and efficiently:

Web Browser: The client-side application is optimized for modern web browsers including Google Chrome, Mozilla Firefox, Microsoft Edge, and Apple Safari.

Backend Framework: The system logic, security validation, and routing are handled by the Laravel PHP Framework.

Database Management System (DBMS): The system will interface with a relational database (e.g., MySQL or PostgreSQL) to safely store, update, and retrieve all business records (Users, Inventory, Logs, Transactions).

Operating System: The web application is OS-agnostic for the end-user. The hosting server will run on a standard web server environment (e.g., Linux/Ubuntu running Nginx or Apache).

5.4. Communication Interfaces

Network Protocol: All data transmissions between the client browsers (cashiers/admins) and the system's server will utilize HTTP/HTTPS protocols. HTTPS (SSL/TLS encryption) must be enforced to protect user login credentials and sensitive financial data from interception.

Connectivity: Client computers must have a stable Local Area Network (LAN) or standard internet connection to communicate with the central database, ensuring real-time stock deductions and preventing concurrent transaction errors.

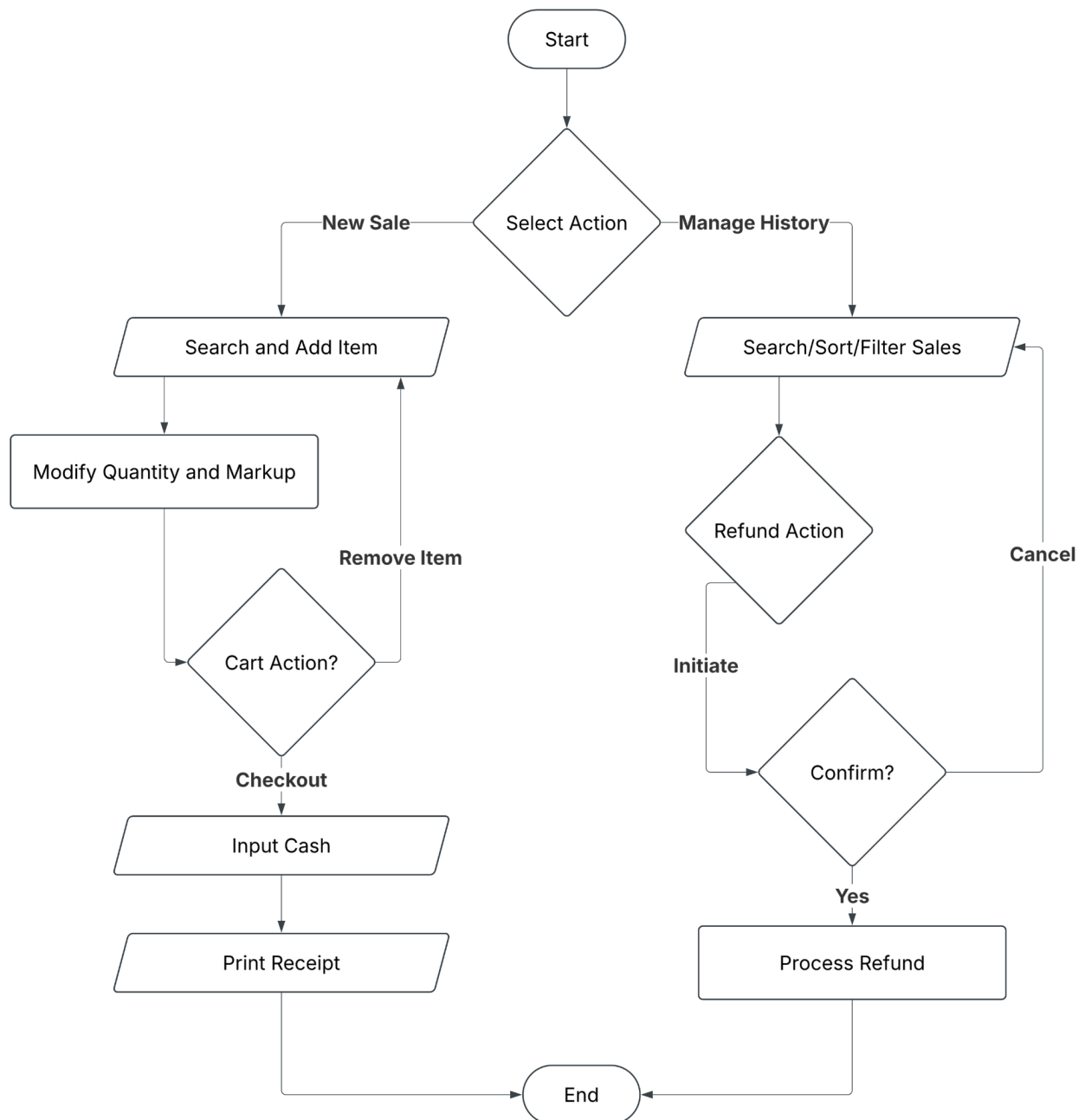
Asynchronous Communication: The frontend will utilize AJAX/Fetch API calls (handled via Alpine.js/JavaScript) to communicate with the Laravel backend for features like live-searching products or triggering toast notifications without requiring full page reloads.

5. APPENDICES

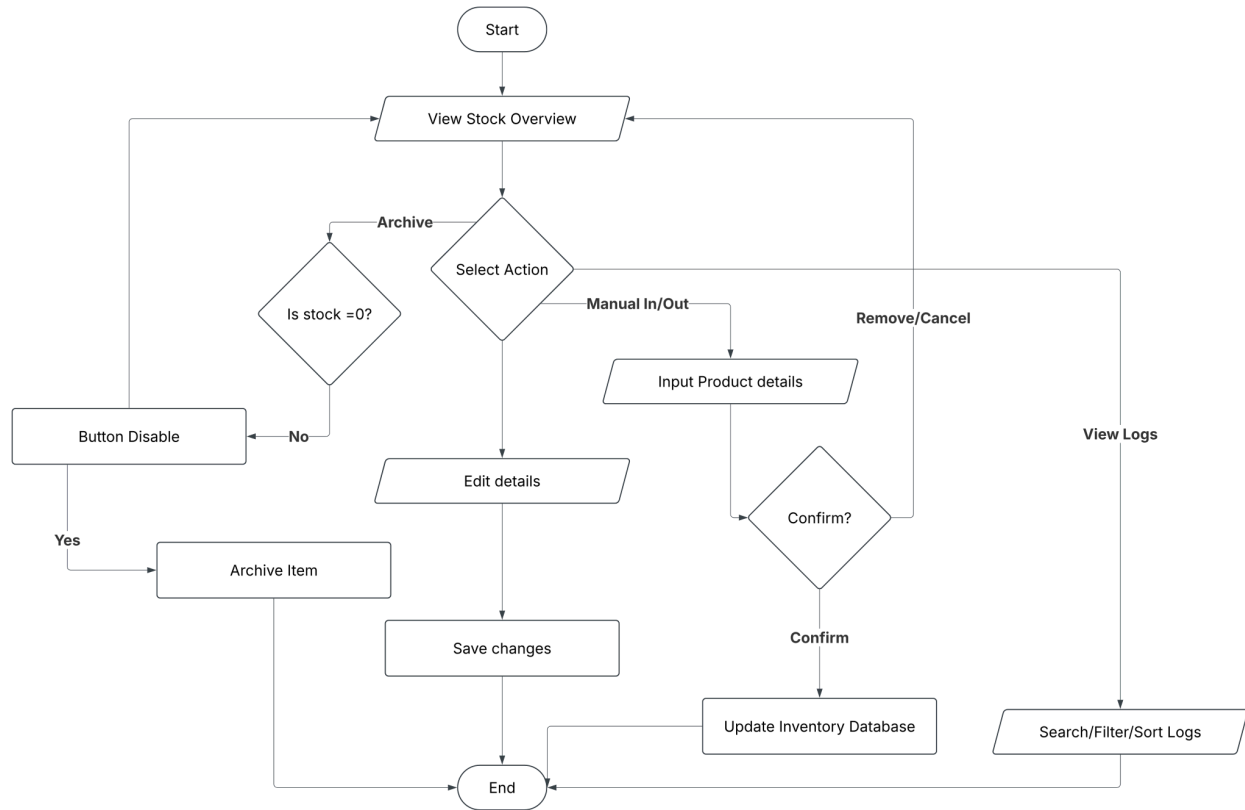
6.1 Detailed Flowchart

The RNM-HMS system is composed of subsystems which are the Point-of-Sales, Inventory Management System, Supplier Records System, Purchasing System, and User and Roles Management System.

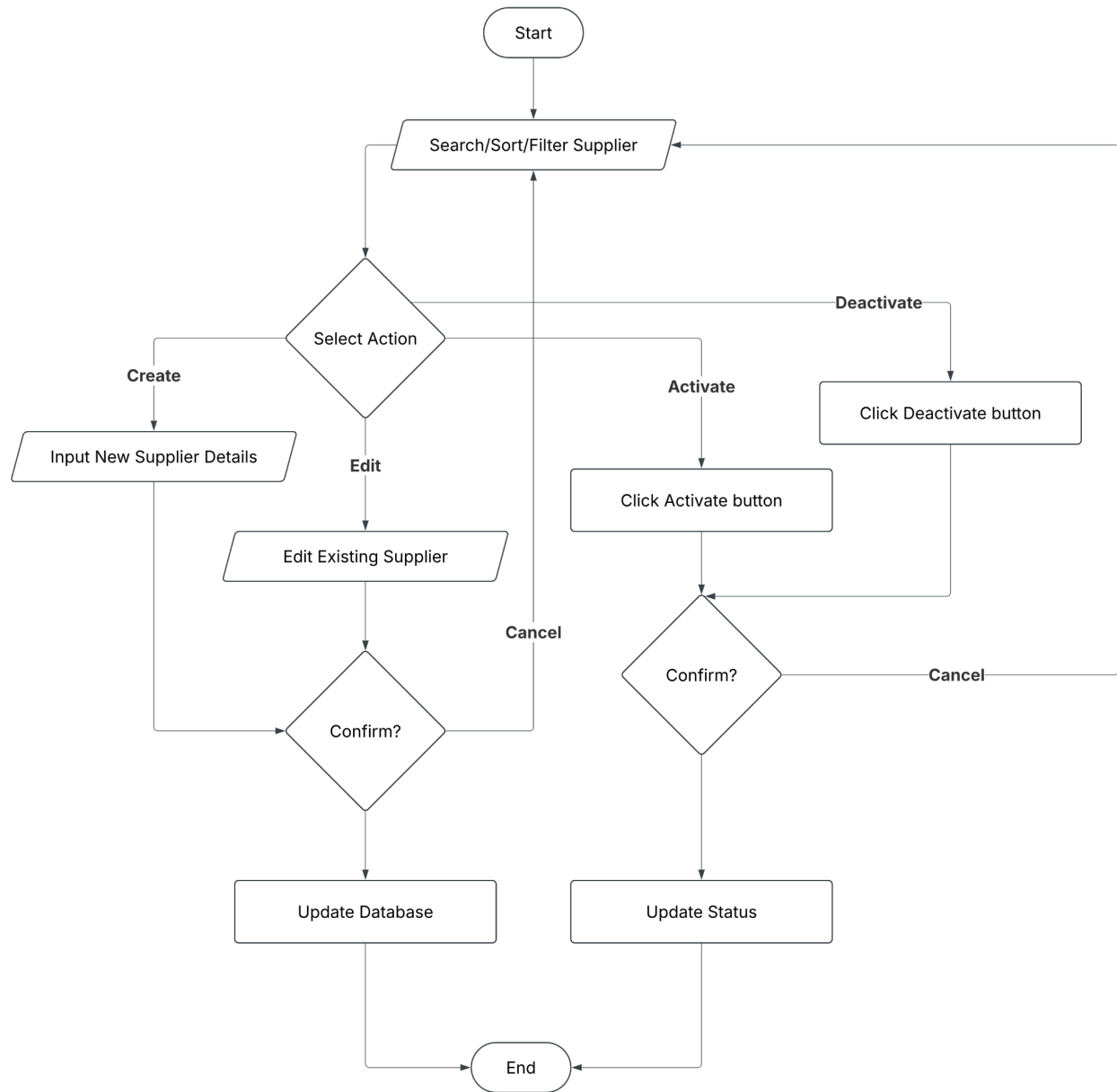
Flowchart for Point of Sales



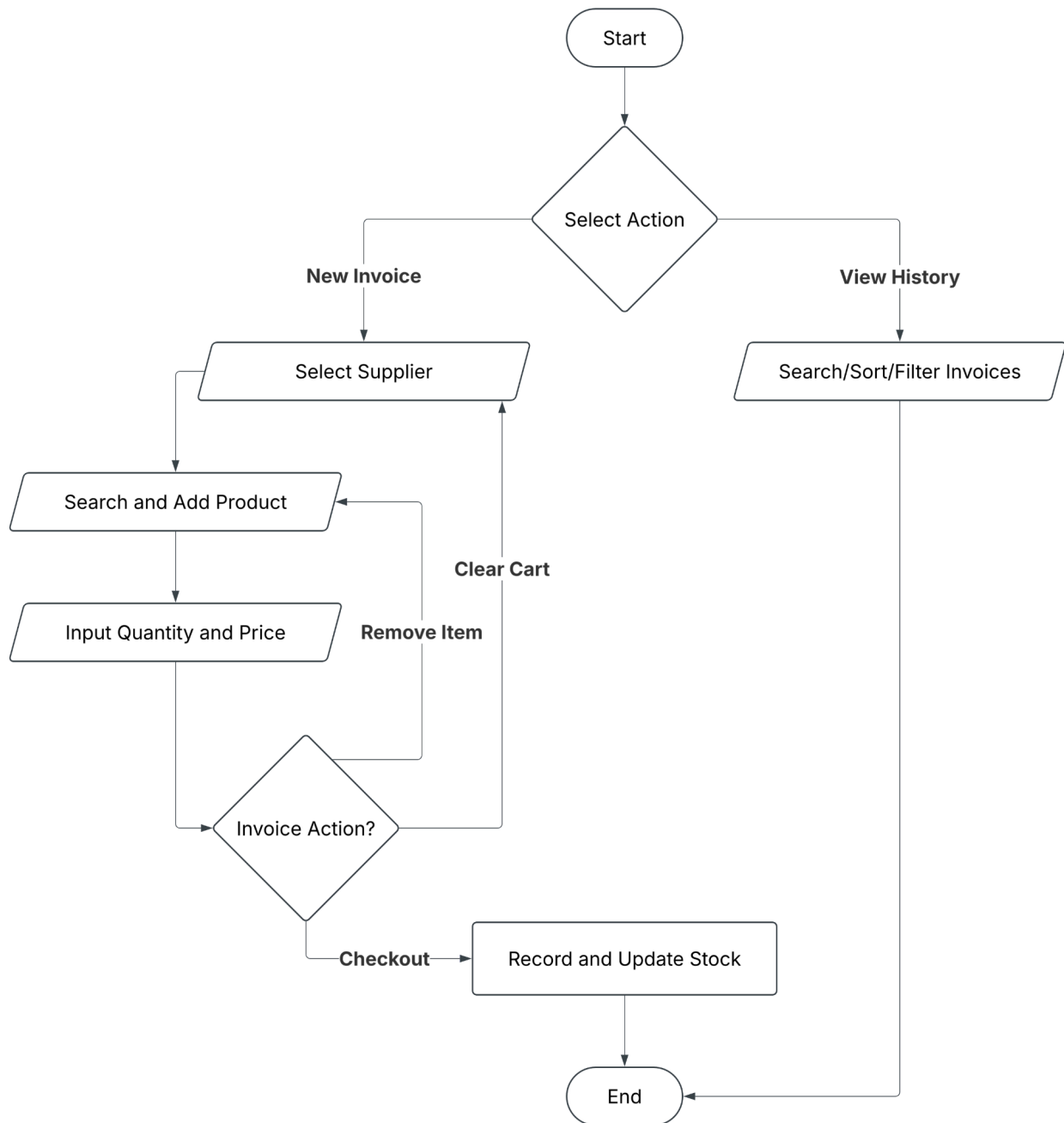
Flowchart for Inventory Management System



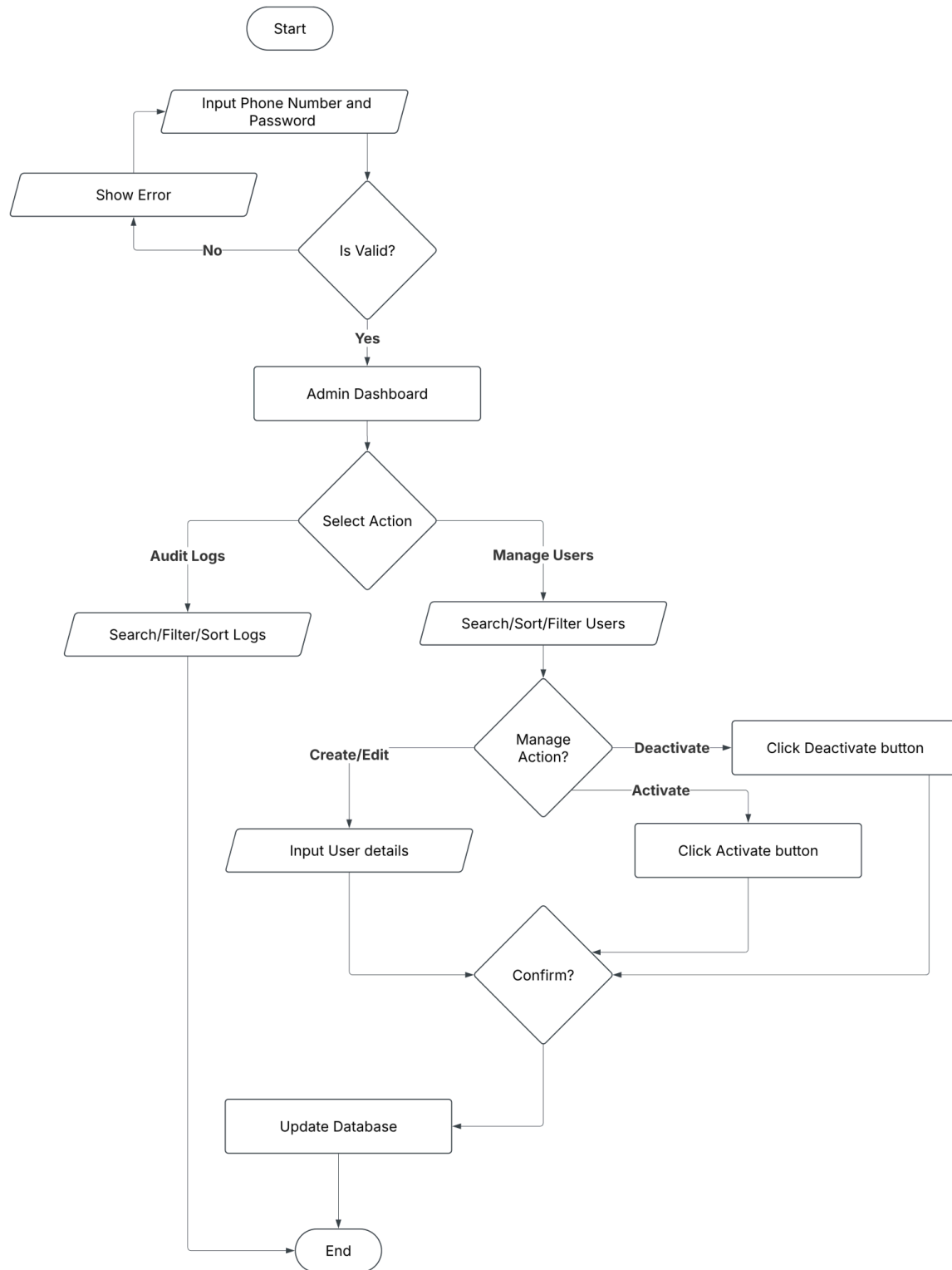
Flowchart for Supplier Records System



Flowchart for Purchasing System



Flowchart for User and Roles Management System



6.2 Data Dictionary

DATA DICTIONARY FOR POINT OF SALES

Entities

Entity Name or Label	Alternative Name(s) or Alias	Description	Input Data Flows	Output Data flows
Admin	N/A	This entity can create new sale, add and remove items to cart, checkout cart, and cancel order.	Create New sale View inventory items Sort inventory items list Search an item Create new cart Modify item markup Remove item to cart Checkout cart Print Receipt Cancel order Refund Sales View sales history Sort sales history list Search a sale history	New sale confirmation Display inventory items Display sorted list Display searched item New cart confirmation Update markup confirmation Delete confirmation Checkout confirmation Print receipt confirmation Cancel confirmation Refund Confirmation Display sales history Display sorted

				list Display search sale history
--	--	--	--	--

Data Flows

Data flow Name or Label	Alternative Name(s) or Alias	Description	Origin	Destination	Volume and Frequency
Create New sale	N/A	Admin can create new sale	Admin	Sales item	Approximat ely 25–30 per day
View inventory items	N/A	Admin can view inventory items	Admin	Products	Approximat ely 25–30 per day
Sort inventory items list	N/A	Admin can sort inventory items	Admin	Products	Approximat ely 10–15 per day
Search an items	N/A	Admin can search Inventory items through code, id, or name	Admin	Products	Approximat ely 25–30 per day
Add items to cart	N/A	Admin can add items to cart	Admin	Sales Item	Approximat ely 25–30 per day
Modify item markup	N/A	Admin can add markup to items	Admin	Product Logs	Approximat ely 5–10 per day
Remove item to cart	N/A	Admin can remove items to	Admin	Sales Item	Approximat ely 5–10 per day

		cart			
Checkout cart	N/A	Admin can checkout cart	Admin	Sales	Approximately 25–30 per day
Print Receipt	N/A	Print receipt of sale	Checkout Cart	Sales	Approximately 15–20 per day
Cancel order	N/A	Admin can cancel order	Admin	Sales Item	Approximately 5–10 per day
View sales history	N/A	Admin can view sales history	Admin	Sales	Approximately 20–25 per day
Sort sales history list	N/A	Admin can sort sales history	Admin	Sales	Approximately 10–15 per day
Search sales history	N/A	Admin can search sales through item name or date	Admin	Sales	Approximately 10–15 per day
Refund Sales	N/A	Admin can refund a sale	Admin	Sales	Approximately 10–15 per day

Data Stores

Data Name Label	Stores or Alternative Name(s) or Alias	Description	Attributes	Volume and Frequency
Products	N/A	Description of the products	ProductId, Name, Capital, unit, QuantityInStock	Approximately 25–30 per day
Sales	N/A	This records	SalesId, Date,	Approximately

		the logs of the sales	UserID, TotalAmount	10–15 per day
SalesItem	N/A	This records the cart details	SalesId, ProductID, Quantity, Markup, Subtotal	Approximately 25–30 per day
ProductLogs	N/A	This records the logs	LogId, ProductId, EmployerId, TransactionType, QuantityChanged, Date, remarks	Approximately 5–10 per day

Processes

Process Number	Process Name	Input Data Flows	Output Data Flows
1.1	Manage Cart	Scanned/Searched Item ID, Item Quantity, Custom Markup Price	Updated Cart Subtotal, Active Cart Items List
1.2	Process Checkout	Final Cart Manifest, Cash Payment Amount	Sales Transaction ID, Stock Deduction Command
1.3	Generate Receipt	Sales Transaction ID, Processed Sales Items	Formatted Receipt Print Data
1.4	Manage Sales History	Search Query, Date Filters, Refund Request	Filtered Sales Record List, Refund Confirmation
1.2	Refund Sales	Sales Transaction ID, Refund Item Details, Original Sales Record	Refund Confirmation Message, Stock Increment Command, Updated Sales

			Ledger Entry, Refund Receipt Details
--	--	--	--

Process Descriptions

Process 1.1 - MANAGE CART

Receive ITEM INFORMATION from ADMIN

Create active checkout session Input values for product quantities and markups

Output updated cart subtotal

Process 1.2 - PROCESS CHECKOUT

Receive PAYMENT INFORMATION from ADMIN

Finalize sale transaction Input cash payment amount

Output sales transaction confirmation and stock deduction

Process 1.3 - GENERATE RECEIPT

Access transaction information from SALES database table

Format data for printing Output physical receipt confirmation

Process 1.4 - MANAGE SALES HISTORY

Receive SEARCH OR REFUND COMMAND from ADMIN

Look up or refund historical transactions

Input search filters or refund request

Output filtered sales list or refund confirmation

Process 1.5 - PROCESS REFUND

Receive TRANSACTION INFORMATION from ADMIN

Process transaction refund

Input values for transaction ID and refunded items

Output refund confirmation and stock adjustment

DATA DICTIONARY FOR INVENTORY MANAGEMENT SYSTEM

Entities

Entity Name or Label	Alternative Name(s) or Alias	Description	Input Data Flows	Output Data flows
Admin	N/A	This entity can view, modify, archive, and stock out items.	View inventory items Sort inventory item lists Search an item Modify existing item Archive item View item logs Sort item logs Filter item logs Search an item logs Stock out item	Display inventory items Display sorted list Display search item Update item confirmation Archive confirmation Display item logs Display sorted logs Display Filtered logs Display searched logs Item stock out confirmation

Data Flows

Data flow Name or Label	Alternative Name(s) or Alias	Description	Origin	Destination	Volume and Frequency
View	N/A	Admin can	Admin	Products	Approximat

inventory items		view inventory items			ely 15-30 per day
Sort inventory item lists	N/A	Admin can sort inventory items	Admin	Products	Approximately 15-30 per day
Search an item	N/A	Admin can search inventory items through code, id or name	Admin	Products	Approximately 15-30 per day
Modify existing item	N/A	Admin can modify existing inventory items	Admin	Products	Approximately 1-15 per day
Archive item	N/A	Admin can archive items	Admin	Archive Products	Approximately 1-15 per day
View item logs	N/A	Admin can view inventory items logs	Admin	Product Logs	Approximately 15-30 per day
Sort item logs	N/A	Admin can sort inventory items logs	Admin	Product Logs	Approximately 15-30 per day
Filter item logs	N/A	Admin can filter items logs	Admin	Product Logs	Approximately 15-30 per day
Search an item logs	N/A	Admin can search inventory items logs through	Admin	Products	Approximately 15-30 per day

		code, id, or name			
Stock out item	N/A	Admin can out stock items	Admin	Products	Approximately 5-15 per day

Data Stores

Data Stores or Label	Alternative Name(s) or Alias	Description	Attributes	Volume and Frequency
Products	N/A	Description of the products	ProductId, Name, Capital, unit, QuantityInStock	Approximately 15-30 per day
Product Logs	N/A	This records the product logs	LogId, ProductId, EmployerId, TransactionType, QuantityChanged, Date, remarks	Approximately 15-30 per day
Archive Logs	N/A	This records the logs of archived items	ProductId, Name, Capital, Unit, QuantityInStock, ArchiveAt	Approximately 1-15 per day

Processes

Process Number	Process Name	Input Data Flows	Output Data Flows
2.1	Monitor Stock Levels	Current QuantityInStock, Minimum Stock Thresholds	Low-Stock Dashboard Alerts

2.2	Manual Stock Adjustment	Product ID, Stock In/Out Quantity, Adjustment Remarks	Updated QuantityInStock, Manual Stock Log Entry
2.3	Archive Zero-Stock Items	Product ID (where Stock = 0)	Archived Status Flag, Archive Log Entry
2.4	Log Stock Movements	Adjustment Data (Sale, Purchase, Manual, Archive)	Saved Movement Ledger Record

Process Descriptions

Process 2.1 - MONITOR STOCK LEVELS

Access stock information from PRODUCTS database table

Evaluate current stock against minimum thresholds

Output low-stock dashboard alerts

Process 2.2 - MANUAL STOCK ADJUSTMENT

Receive ADJUSTMENT INFORMATION from ADMIN

Update product stock levels manually

Input values for stock in/out quantities and remarks

Output stock adjustment confirmation

Process 2.3 - ARCHIVE ZERO-STOCK ITEMS

Receive ARCHIVE COMMAND from ADMIN

Migrate active product to archive Input zero-stock product reference

Output archived status confirmation

Process 2.4 - LOG STOCK MOVEMENTS

Receive MOVEMENT DATA from INVENTORY MODULE

Record stock adjustment history

Update product logs data store

Output movement ledger record

DATA DICTIONARY FOR SUPPLIER RECORDS MANAGEMENT

Entities

Entity Name or Label	Alternative Name(s) or Alias	Description	Input Data Flows	Output Data flows
Admin	N/A	This entity can add, modify, and delete supplier detail	Create new supplier profile Update supplier information Deactivate supplier profile Activate supplier profile Sort supplier list Search a supplier profile Views supplier profiles	New supplier confirmation Update supplier information detail confirmation Display sorted supplier list Deactivated supplier profile Activated supplier profile Display searched supplier profile Display supplier profile

Data Flows

Data flow Name or Label	Alternative Name(s) or Alias	Description	Origin	Destination	Volume and Frequency
Create new supplier	N/A	Admin can create new supplier profile	Admin	Suppliers	Approximately 1-5 per year
Update supplier information	N/A	Admin can modify existing supplier	Admin	Supplier logs	Approximately 1-5 per month

		information			
Delete supplier profile	N/A	Admin can delete existing supplier profile	Admin	Suppliers	Approximately 1-5 per year
Sort supplier list	N/A	Admin can sort existing supplier profiles	Admin	Suppliers	Approximately 10-15 per day
Search supplier profile	N/A	Admin can search existing supplier profile	Admin	Suppliers	Approximately 10-15 per day
View supplier profiles	N/A	Admin can view existing supplier profiles	Admin	Suppliers	Approximately 10-15 per day

Data Stores

Data Stores or Name Label	Alternative Name(s) or Alias	Description	Attributes	Volume and Frequency
Suppliers	N/A	The personal information of suppliers	SupplierId, SupplierName, ContactPerson, CompanyAddress, ContactNumber	Approximately 1-5 per year
Supplier Logs	N/A	This records the logs of a supplier information	Id,SupplierId, OldContactPerson, NewContactPerson, OldContactNu	Approximately 10-15 per day

			mber, NewContactN umber, DateTime, EmployerId	
--	--	--	---	--

Processes

Process Number	Process Name	Input Data Flows	Output Data Flows
3.1	Register Supplier	Supplier Name, Contact Person, Address, Phone Number	New Supplier Profile Record
3.2	Update Supplier Profile	Supplier ID, Modified Contact Details	Updated Supplier Profile, Supplier Change Log
3.3	Toggle Supplier Status	Supplier ID, Status Toggle Command	Updated Supplier Status Flag (Active/Inactive)

Process Descriptions

Process 3.1 - REGISTER SUPPLIER

Receive SUPPLIER INFORMATION from ADMIN

Create new supplier profile Input values for supplier contact fields

Output new supplier confirmation

Process 3.2 - UPDATE SUPPLIER PROFILE

Receive MODIFIED INFORMATION from ADMIN

Update existing supplier details Input new values for supplier fields

Output supplier update confirmation

Process 3.3 - TOGGLE SUPPLIER STATUS

Receive STATUS COMMAND from ADMIN

Change supplier status to active or inactive

Update supplier data store

Output supplier status confirmation

DATA DICTIONARY FOR PURCHASING SYSTEM

Entities

Entity Name or Label	Alternative Name(s) or Alias	Description	Input Data Flows	Output Data flows
Admin	N/A	This entity can choose supplier, create new invoice and view invoice history	Choose a supplier Add existing stocks Search a supplier Sort supplier list Create an invoice Add items to invoice Search an item Sort item list Create new item View invoice history	Supplier confirmation Display supplier Display sorted list New invoice confirmation Update invoice Display searched item Display sorted list New item confirmation Display invoice history

Data Flows

Data flow Name or Label	Alternative Name(s) or Alias	Description	Origin	Destination	Volume and Frequency
Choose supplier	N/A	Admin can choose	Admin	Supplier	Approximately 25–30

		supplier			per day
Add existing stocks	N/A	Admin can add existing items	Admin	Products	Approximately 1-5 per month
Display searched supplier	N/A	Admin can search suppliers	Admin	Suppliers	Approximately 25–30 per day
Display sorted list	N/A	Admin can sort suppliers	Admin	Suppliers	Approximately 25–30 per day
Create invoice	N/A	Admin can create new invoice	Admin	Invoices	Approximately 5–10 per day
Add items to invoice	N/A	Admin can add items to the invoice	Admin	Invoices	Approximately 5–10 per day
Display search item	N/A	Admin can search items to be added	Admin	Products	Approximately 25–30 per day
Display sorted list	N/A	Admin can sort items to be added	Admin	Products	Approximately 25–30 per day
Create new item	N/A	Save invoice to system	Admin	Products	Approximately 5–15 per day
Display invoice history	N/A	Admin can view invoice history	Admin	Invoices	Approximately 25–30 per day

Data Stores

Data Name Label	Stores or	Alternative Name(s) or Alias	Description	Attributes	Volume and Frequency
Invoices		N/A	This records the invoice information	InvoiceId, PurchaseId, DateIssued, TotalAmount, DueDate	
Suppliers		N/A	The personal information of suppliers	SupplierId, SupplierName, ContactPerson, CompanyAddress, ContactNumber	Approximately 25–30 per day
Products		N/A	Description of the products	ProductId, Name, Capital, unit, QuantityInStock	Approximately 5–15 per day

Processes

Process Number	Process Name	Input Data Flows	Output Data Flows
4.1	Create Purchase Invoice	Selected Supplier ID, Invoice Date, Purchase Reference	New Purchase Invoice Header
4.2	Register New Product	New Hardware Details (Name, Unit, Capital Cost)	New Product Record (sent to IMS)
4.3	Process Procurement Checkout	List of Purchased Items, Delivered Quantities, Total	Saved Invoice Record, Stock Addition Command

		Cost	
--	--	------	--

Process Descriptions

Process 4.1 - CREATE PURCHASE INVOICE

Receive INVOICE INFORMATION from ADMIN

Create new inbound delivery manifest

Input values for supplier and invoice date

Output purchase invoice header confirmation

Process 4.2 - REGISTER NEW PRODUCT

Receive NEW HARDWARE INFORMATION from ADMIN

Create new inventory item from purchasing

Input values for product name, unit, and capital cost

Output new product confirmation

Process 4.3 - PROCESS PROCUREMENT CHECKOUT

Receive DELIVERY INFORMATION from ADMIN

Finalize wholesale purchase and increase stock

Input received items and total cost

Output invoice confirmation and stock addition confirmation

DATA DICTIONARY FOR USER MANAGEMENT SYSTEM

Entities

Entity Name or Label	Alternative Name(s) or Alias	Description	Input Data Flows	Output Data flows
Admin	N/A	This entity can create, update and archive the accounts.	Create new account Update/archive existing account	New account confirmation Update/Archive account confirmation

			Display account Filter/Sort/search account	View account list Filter/Sort/Search account list
--	--	--	---	--

Data Flows

Data flow Name or Label	Alternative Name(s) or Alias	Description	Origin	Destination	Volume and Frequency
Create a new account	N/A	Admin creates a new account	Admin	Employers	Approximately 1-3 per year
Display account list	N/A	Admin views accounts	Admin	Employers	Approximately 10-15 per day
Update existing account	N/A	Admin updates existing account information	Admin	Employers	Approximately 1-5 per week
Sort account	N/A	Admin sorts accounts	Admin	Employers	Approximately 10-15 per day
Search Account	N/A	Admin searches account	Admin	Employers	Approximately 10-15 per day

Delete account	N/A	Admin deletes account	Admin	Employers	Approximately 1-3 per year
----------------	-----	-----------------------	-------	-----------	----------------------------

Data Stores

Data Name Label	Stores or Alternative Name(s) or Alias	Description	Attributes	Volume and Frequency
Employers	N/A	Personal information of the employer	EmployerId, Name, Role, DateAdded, Email	Approximately 1-3 new registered account per year

Processes

Process Number	Process Name	Input Data Flows	Output Data Flows
5.1	Authenticate User	Username, Password/PIN	Session Token, Role Authorization Level
5.2	Manage Employee Accounts	Employee Name, Role, Email, Password, Status Toggle	New/Updated Employer Record
5.3	Track System Audit Logs	User ID, Action Trigger (e.g., "Deleted Item"), Timestamp	Saved System Audit Log Entry

Process Descriptions

Process 5.1 - AUTHENTICATE USER

Receive LOGIN INFORMATION from ADMIN

Verify user credentials Input values for username and password

Output session token and role authorization

Process 5.2 - MANAGE EMPLOYEE ACCOUNTS

Receive ACCOUNT INFORMATION from ADMIN

Create or update employee account

Input values for user role, email, and password

Output employee account confirmation

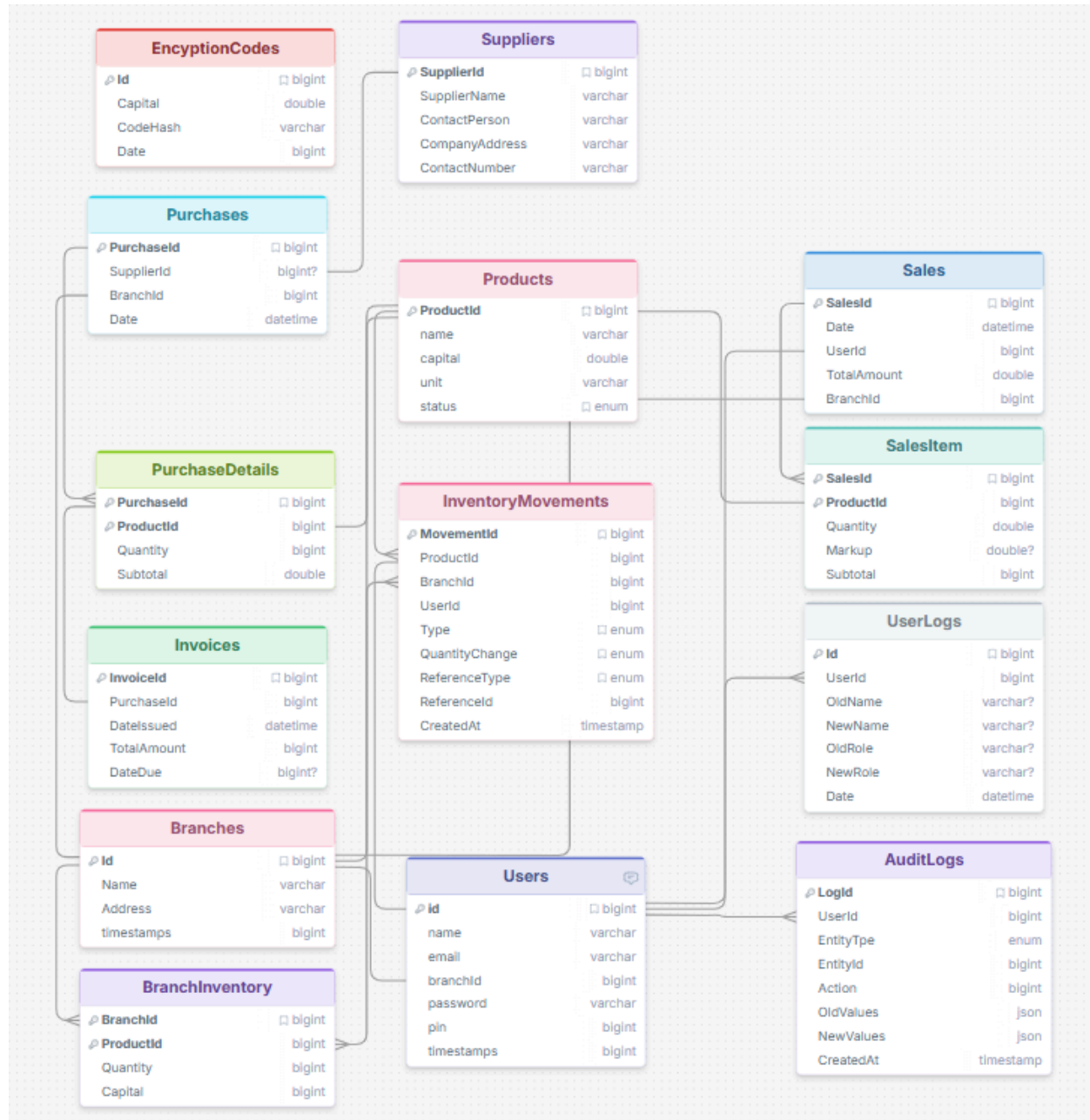
Process 5.3 - TRACK SYSTEM AUDIT LOGS

Receive ACTION INFORMATION from ALL SUBSYSTEMS

Record system-wide user actions Update audit logs data store

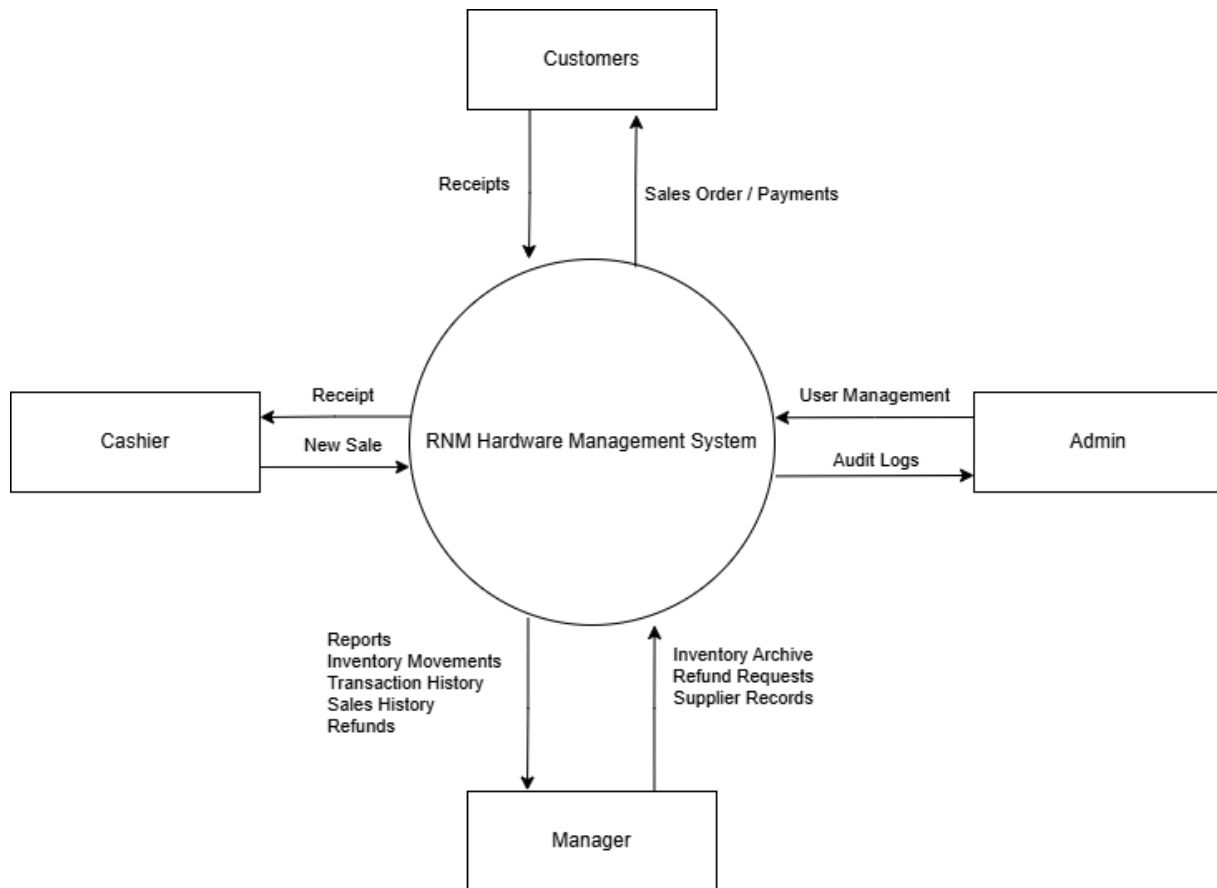
Output system audit log entry

6.3 Entity Relationship Diagram

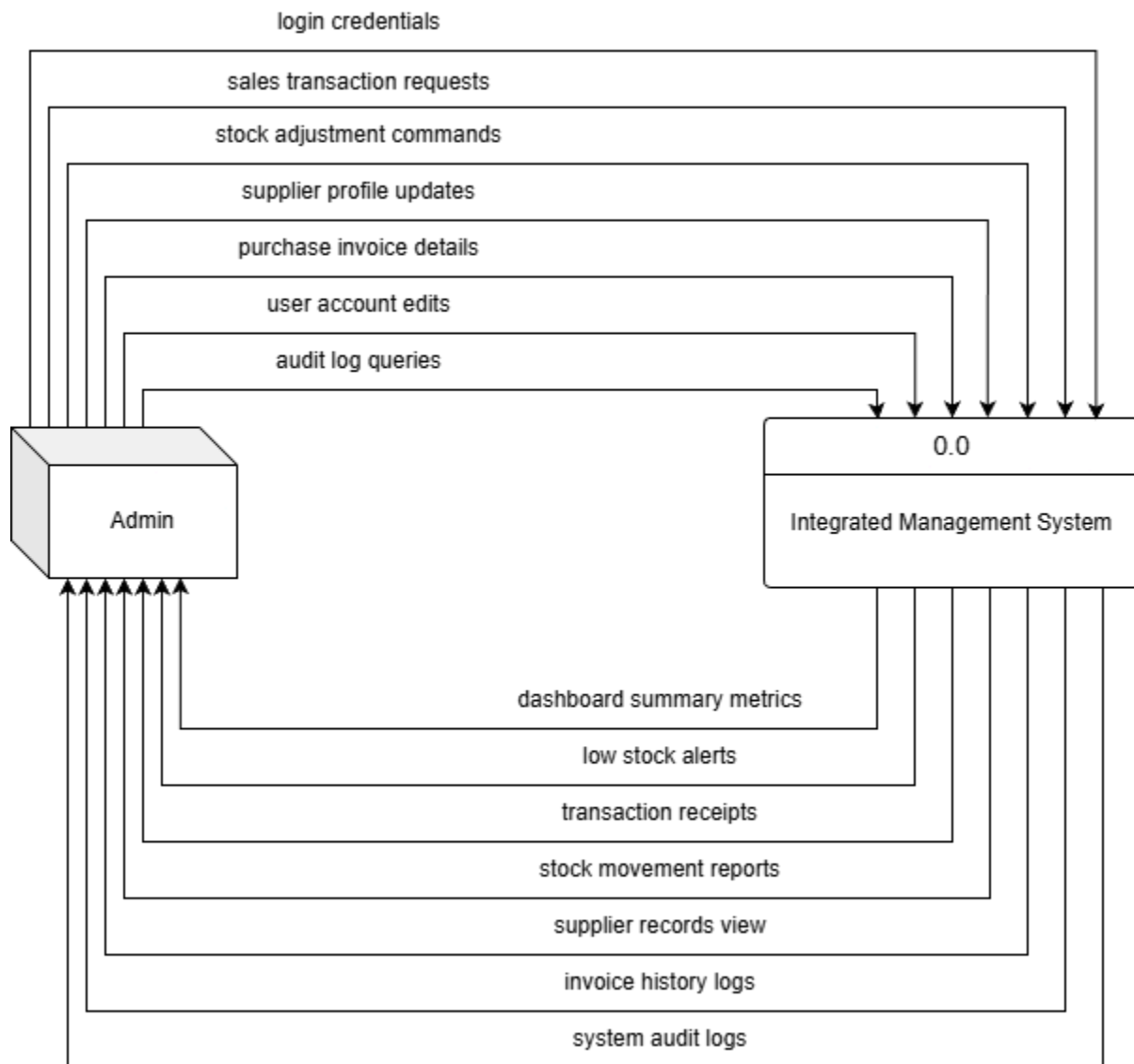


6.4 Data Flow Diagram

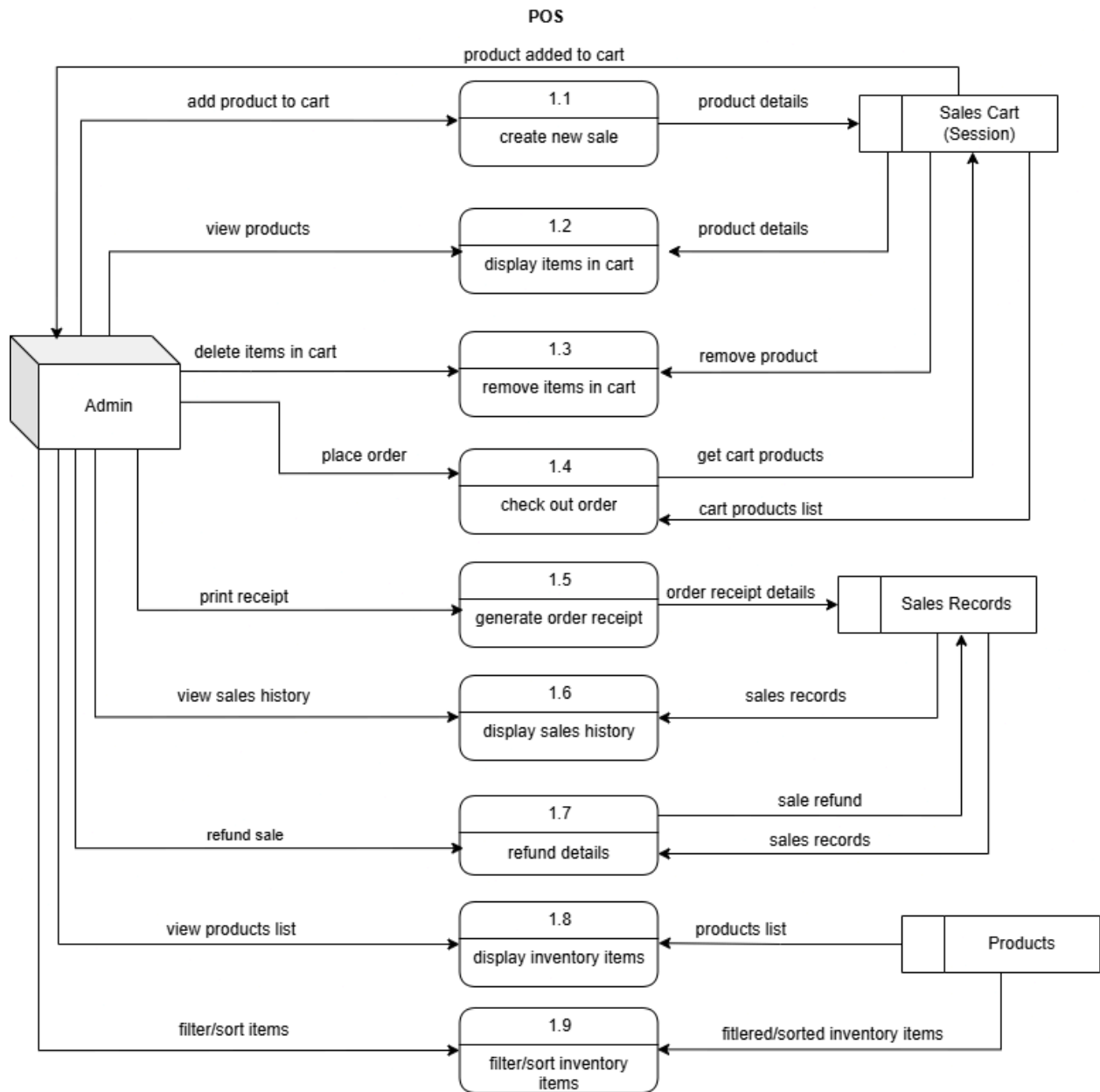
Context Diagram



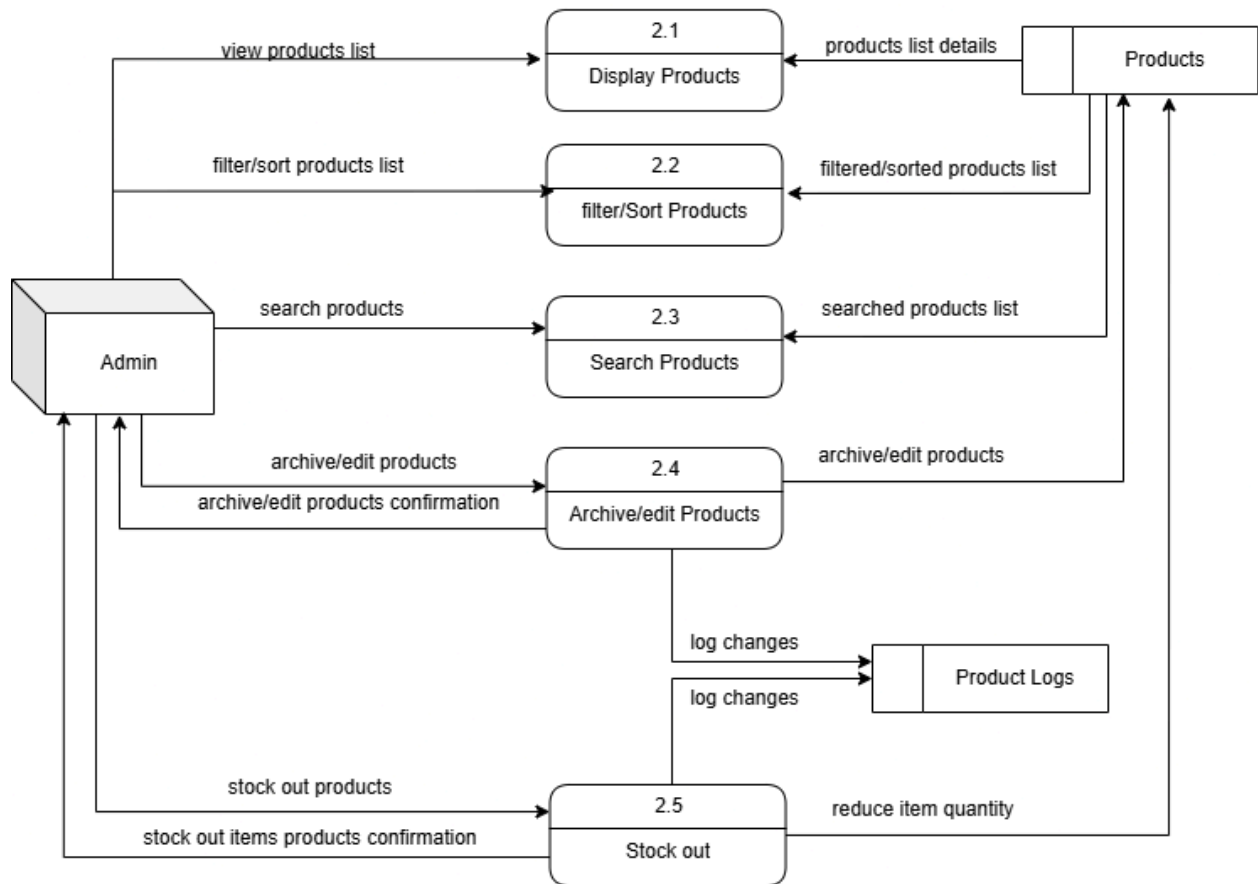
0 - Level Data Flow Diagram



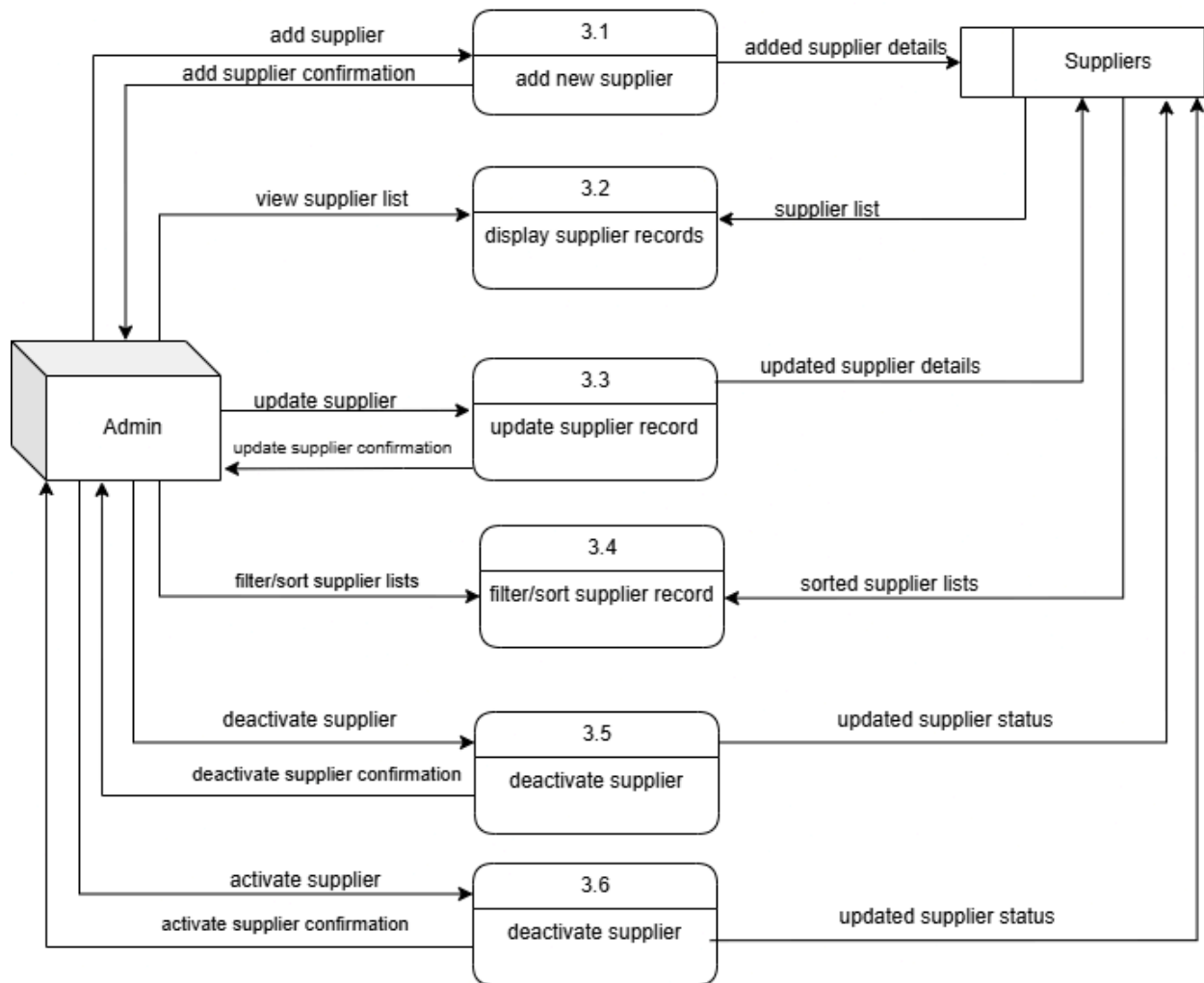
1 - Level Data Flow Diagram



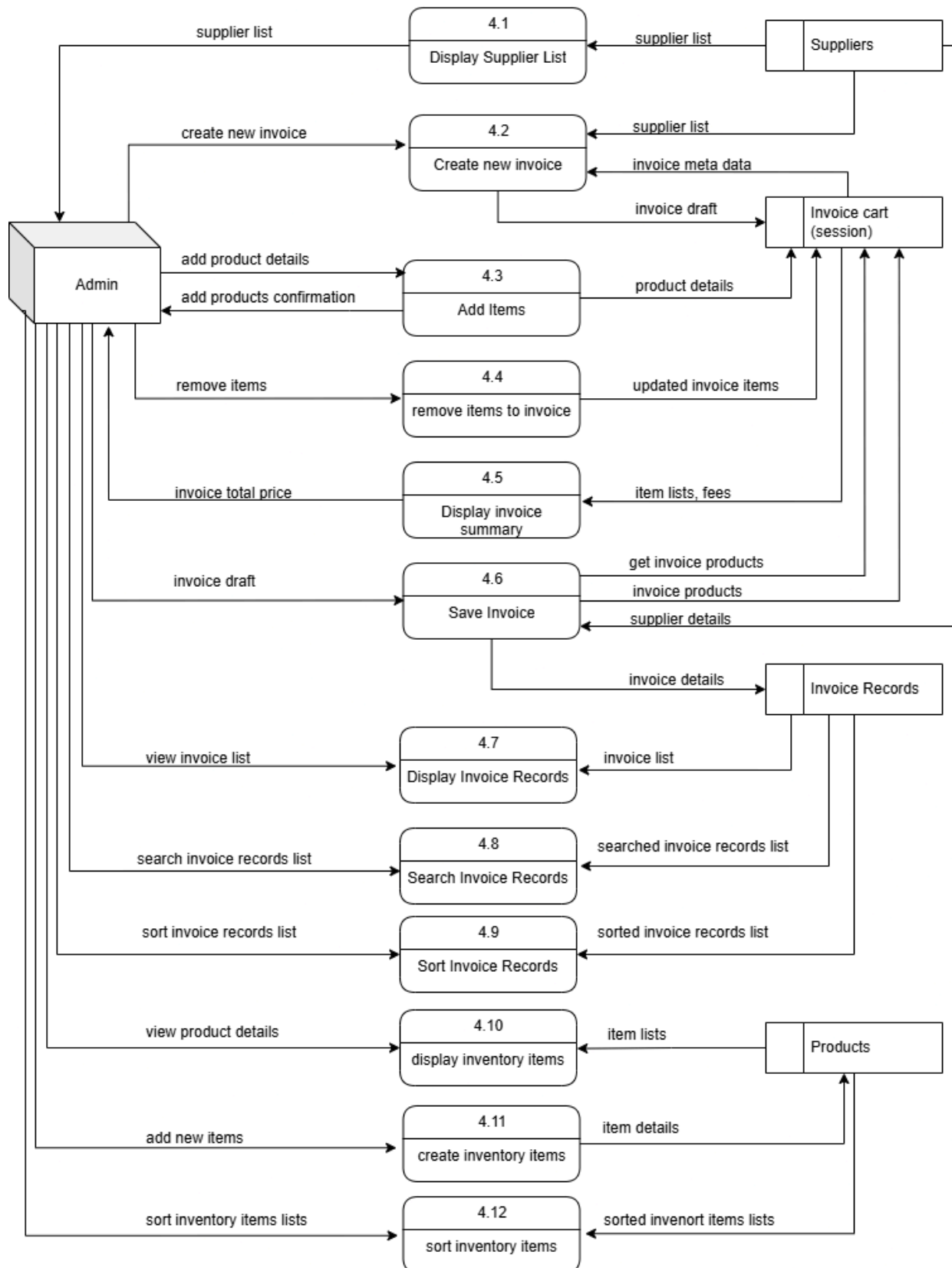
Inventory Management



Supplier Management



Purchasing Management



User Management

